

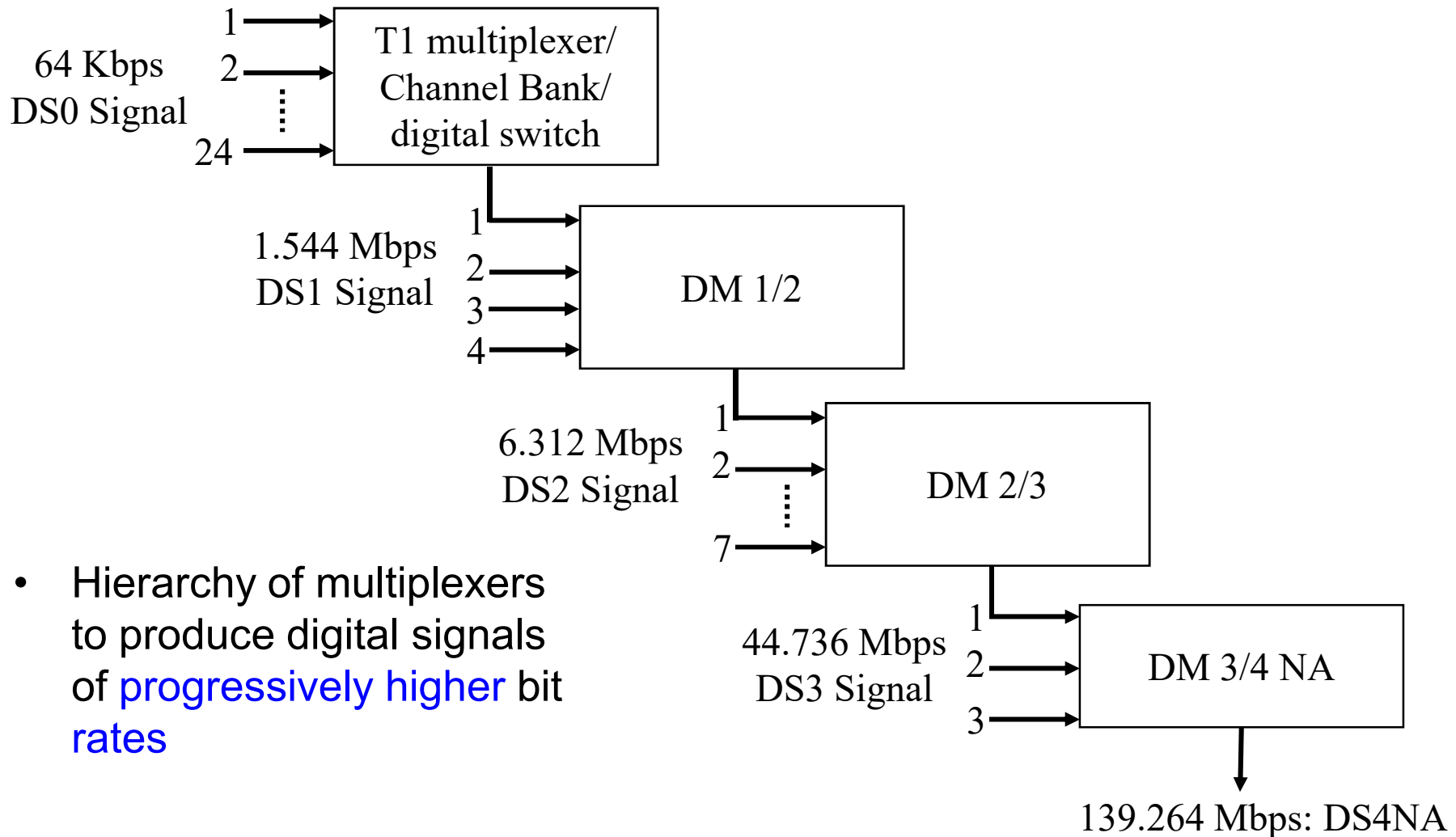
CSE 311:

Data Communication

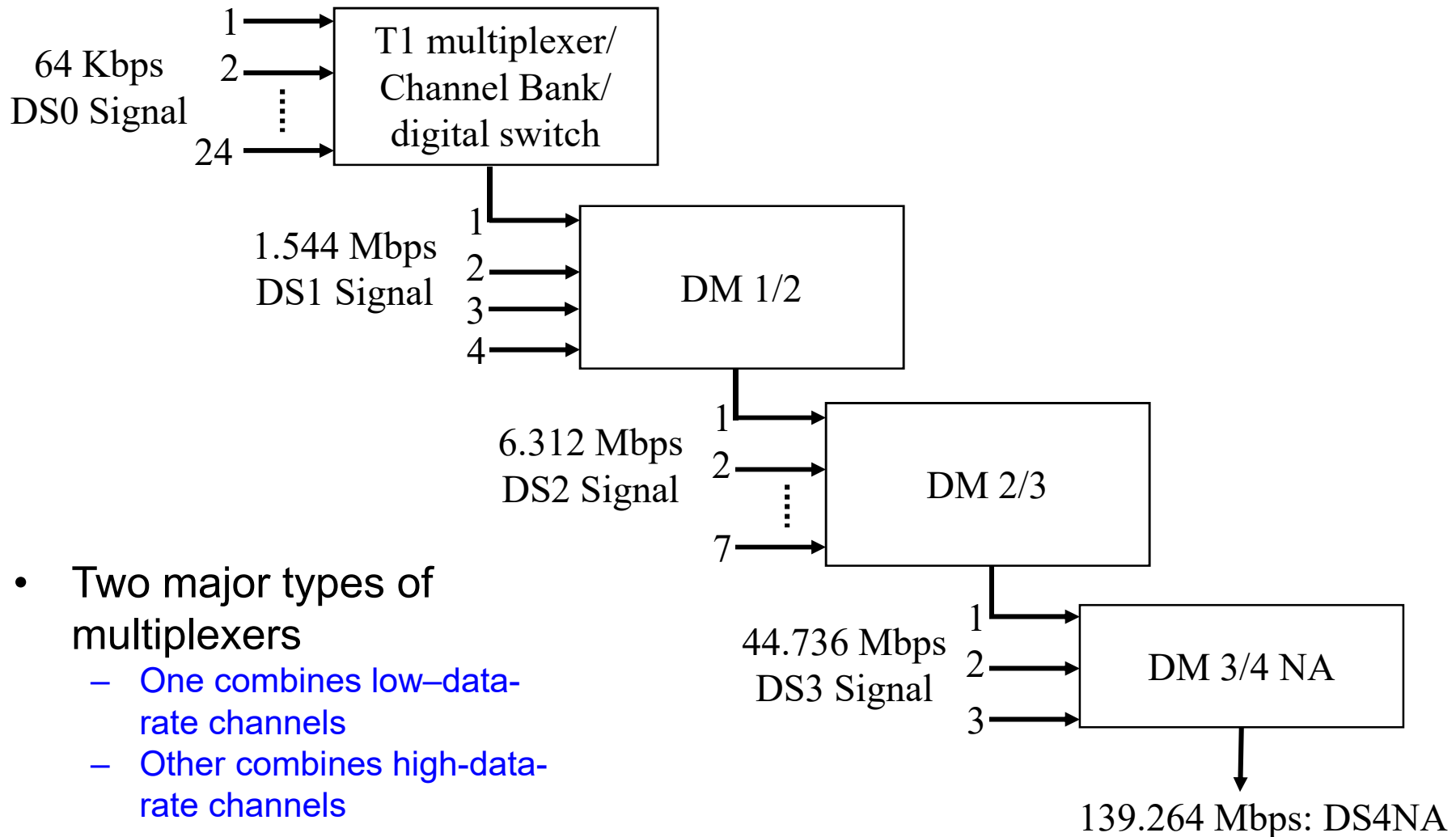
Instructor:
Dr. Md. Monirul Islam

Sampling and Analog-to-Digital Conversion

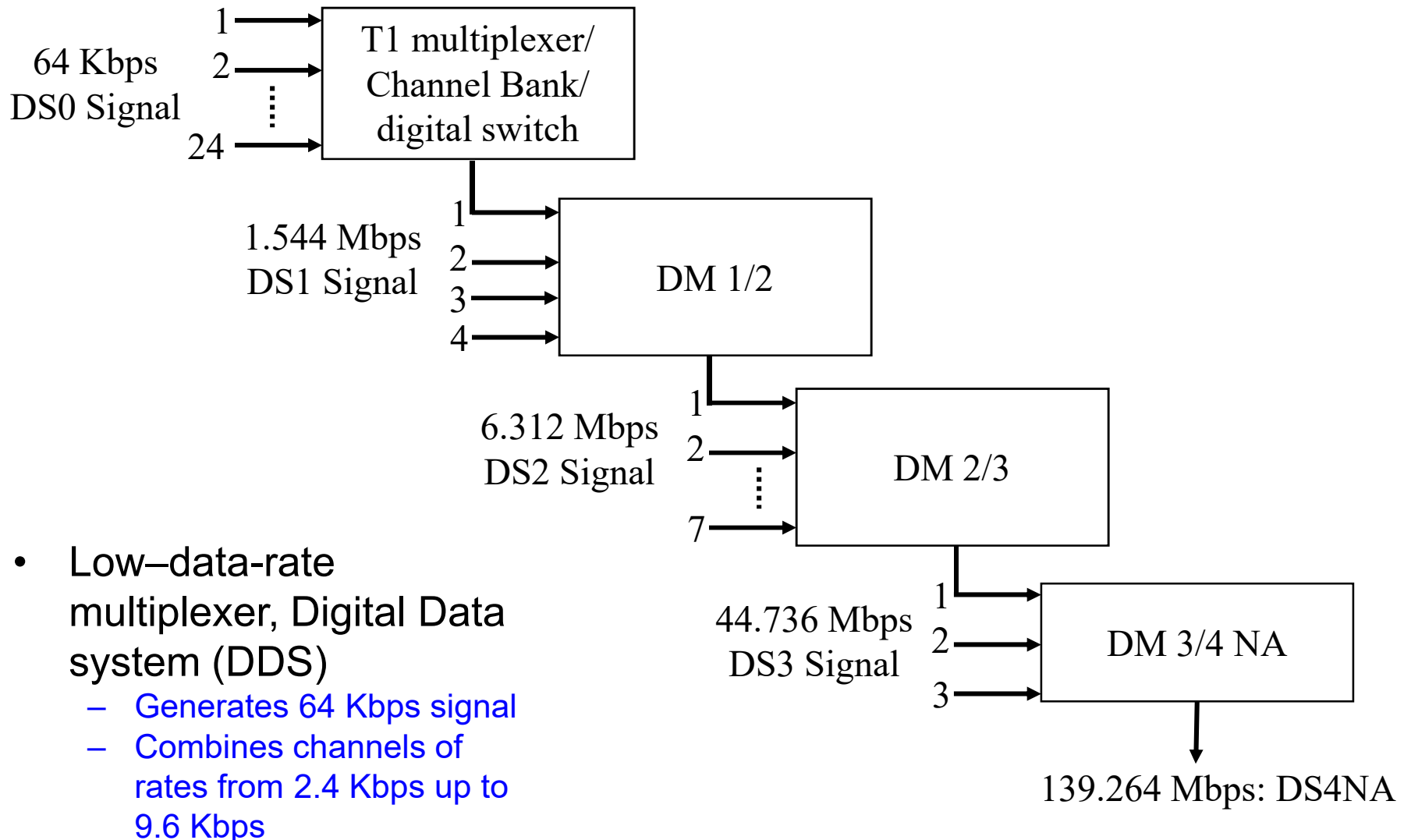
Plesiochronous (almost Synchronous) Digital Hierarchy



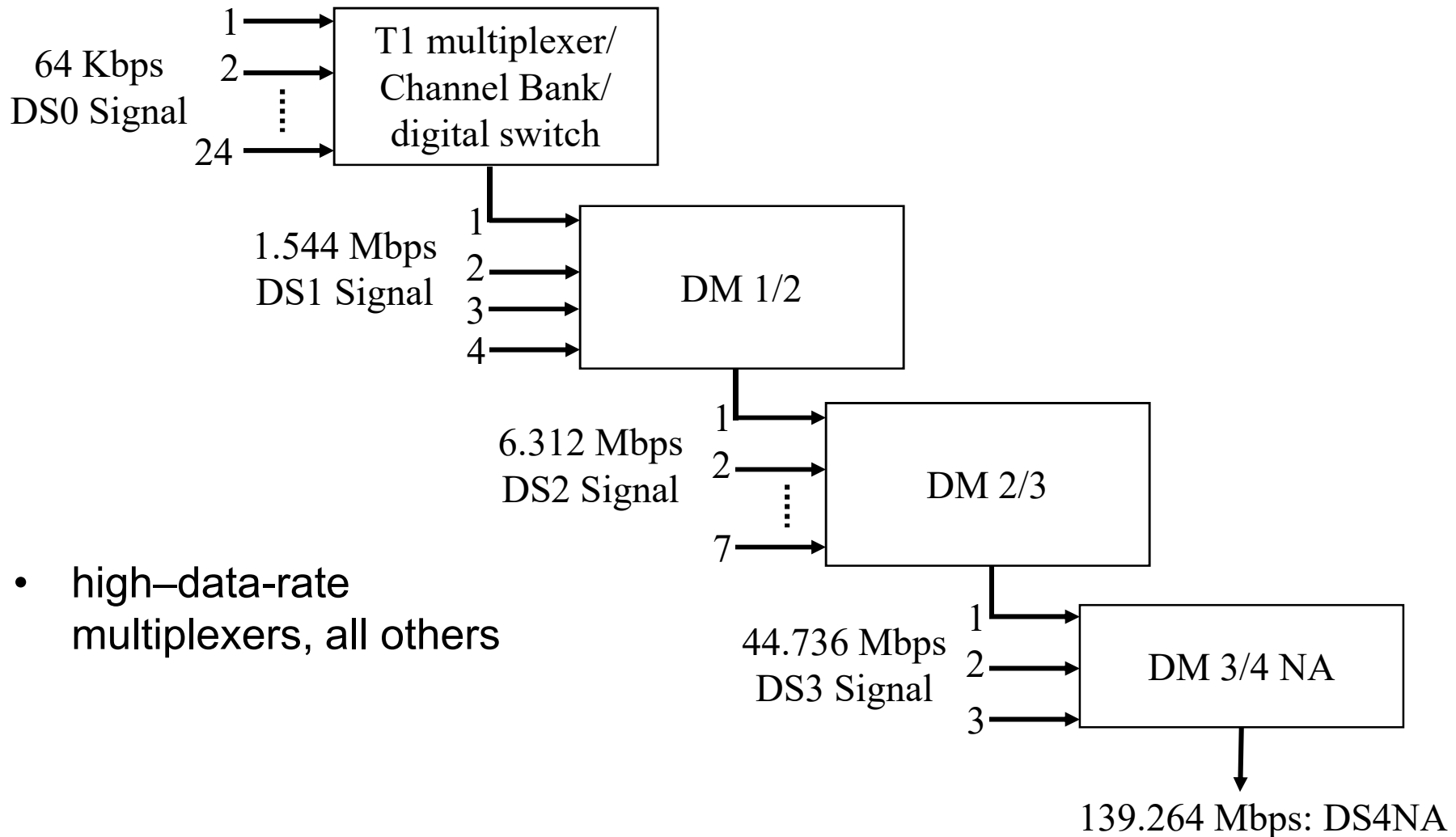
Plesiochronous (almost Synchronous) Digital Hierarchy



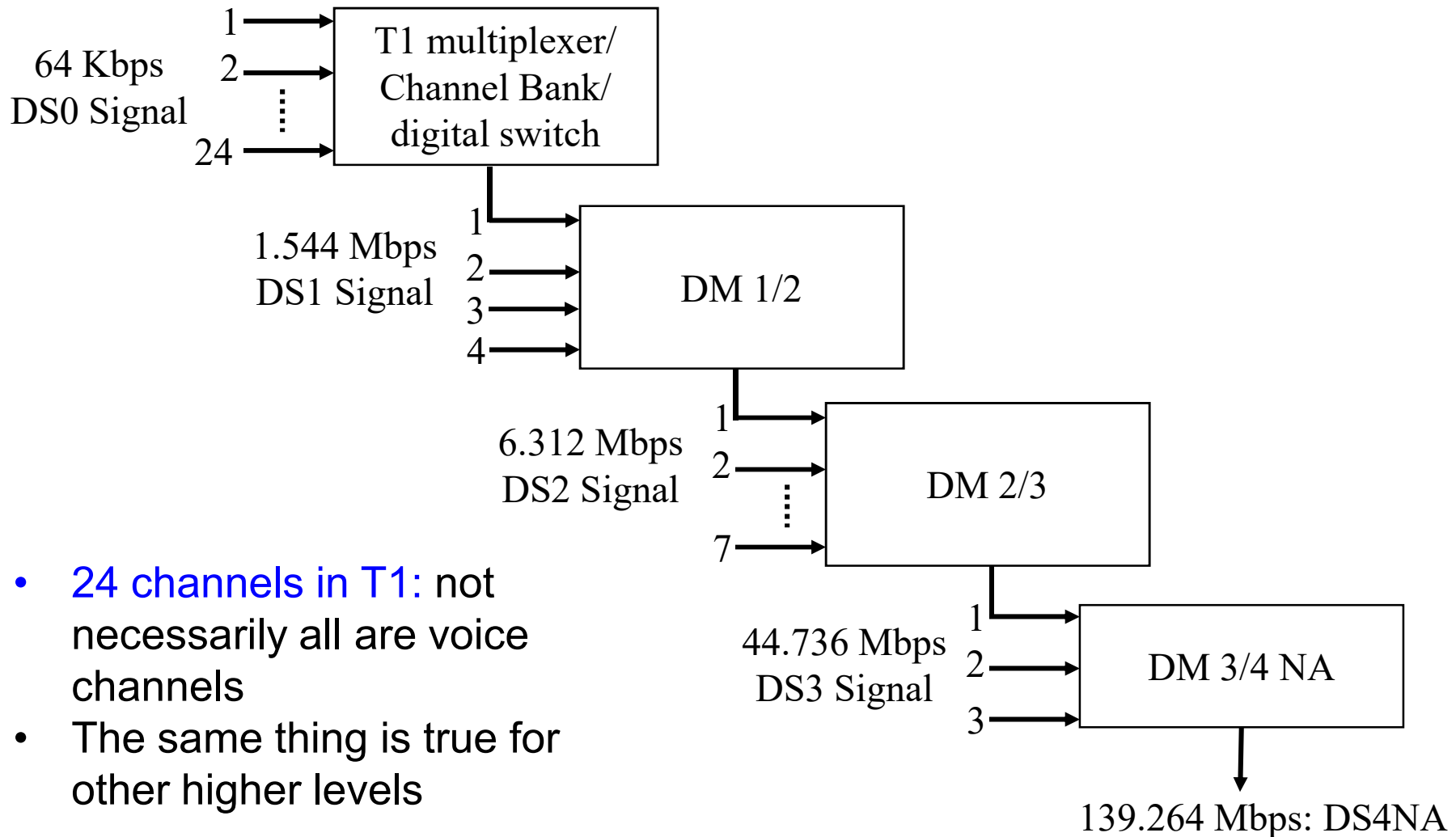
Plesiochronous (almost Synchronous) Digital Hierarchy



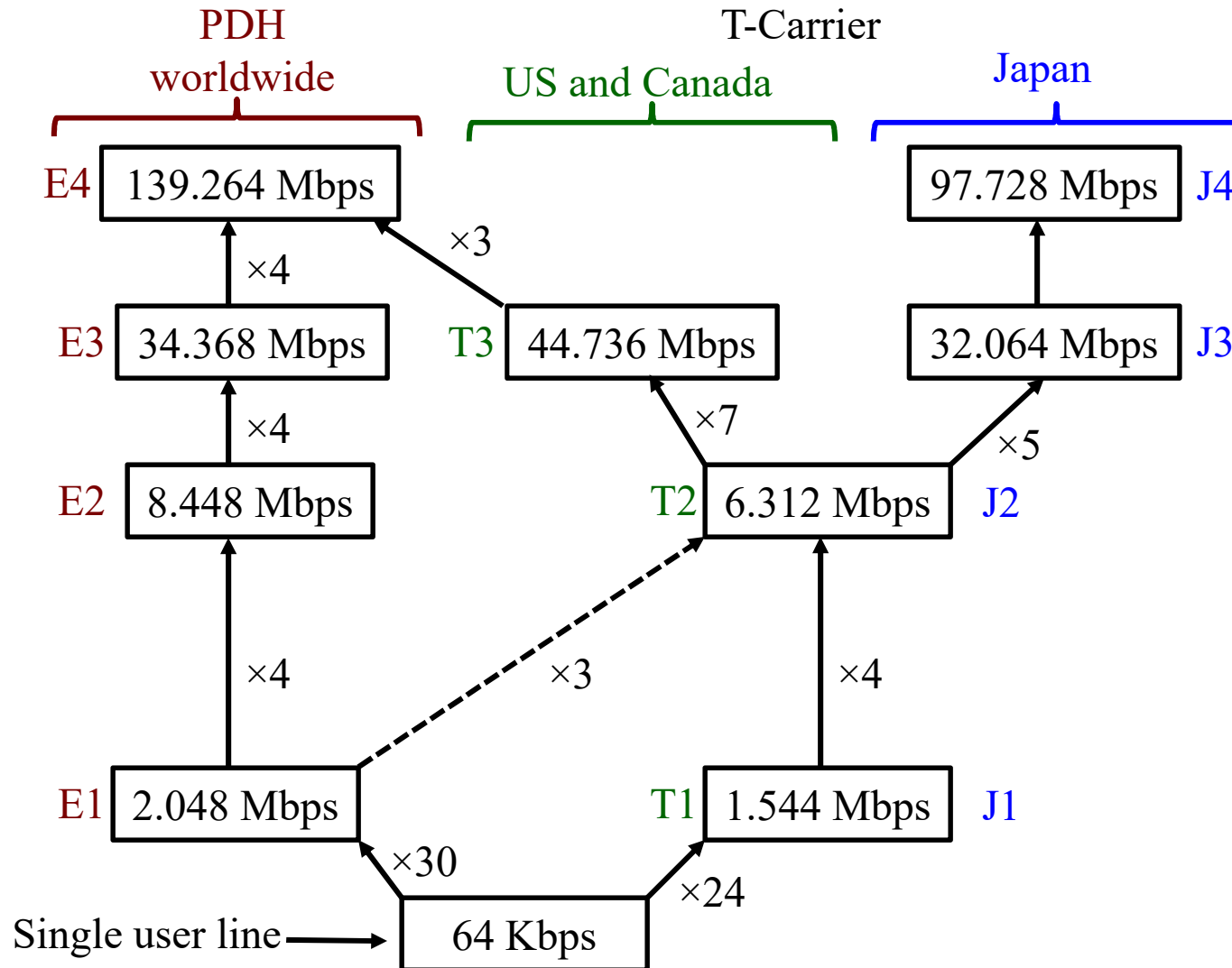
Plesiochronous (almost Synchronous) Digital Hierarchy



Plesiochronous (almost Synchronous) Digital Hierarchy



Other Plesiochronous Digital Hierarchies



Differential Pulse Code Modulation (DPCM)

PCM and Transmission Channel

Bandwidth: Example-1

Analog signal bandwidth $\Rightarrow$ digital data rate $\Rightarrow$ Channel Bandwidth requirement

Audio signal b/w = 15 KHz

However, up to 3400 Hz is sufficient for articulation (intelligibility)

Signal bandwidth, $B = 3400$ Hz

Sampling frequency, $f_s = 8000 > 2B$

PCM Quantization level, $L = 256$ (8 bits)

Data rate = $8000 * 8 = 64000$ pulse/second = 64 Kbps

PCM and Transmission Channel

Bandwidth: Example-2

Review

Example 2: data rate for compact disc

High Fidelity is required!

Audio signal b/w = 20 KHz

$$B = 20000 \text{ Hz}$$

$$f_s = 44100 \text{ Hz} > 2B$$

Quantization level, $L = 65,536$ (16 bits)

$$\text{Data rate} = 44100 * 16 = 1.4 \text{ Mbps} = 1400 \text{ Kbps}$$

PCM and Transmission Channel Bandwidth

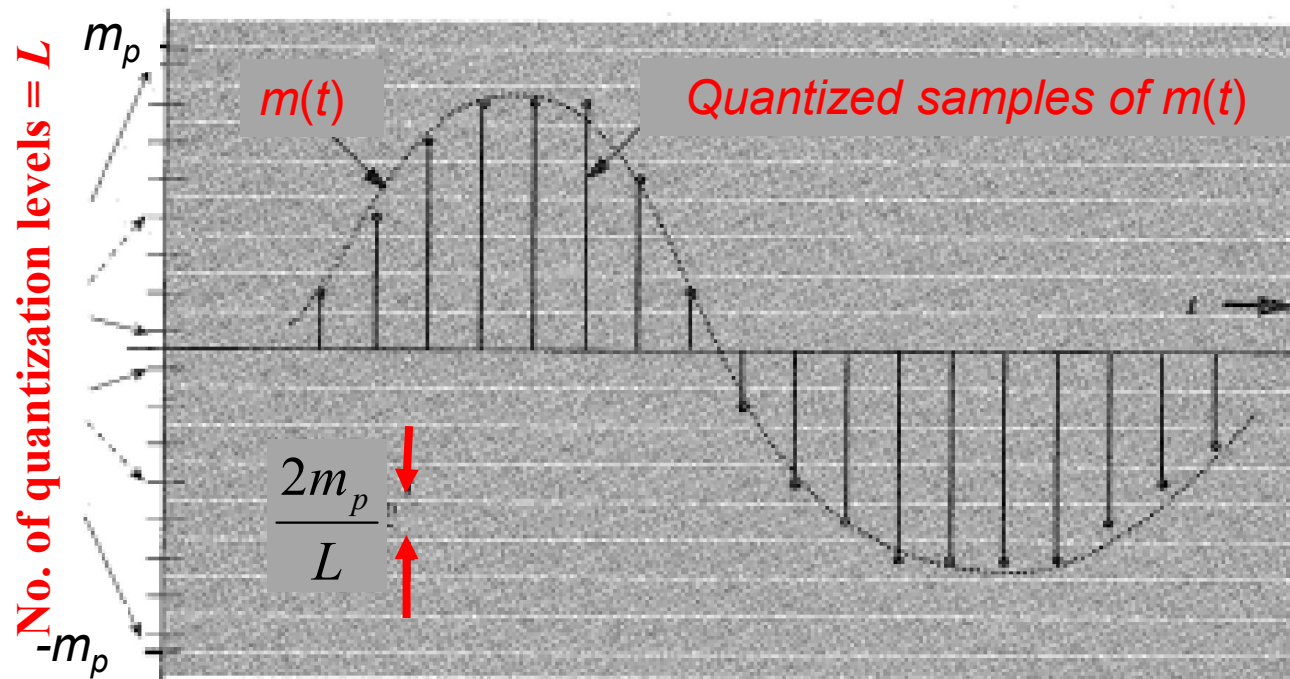
Findings

- PCM generates too many bits
- Requires high bandwidth to transmit them

PCM and Transmission Channel Bandwidth

Causes:

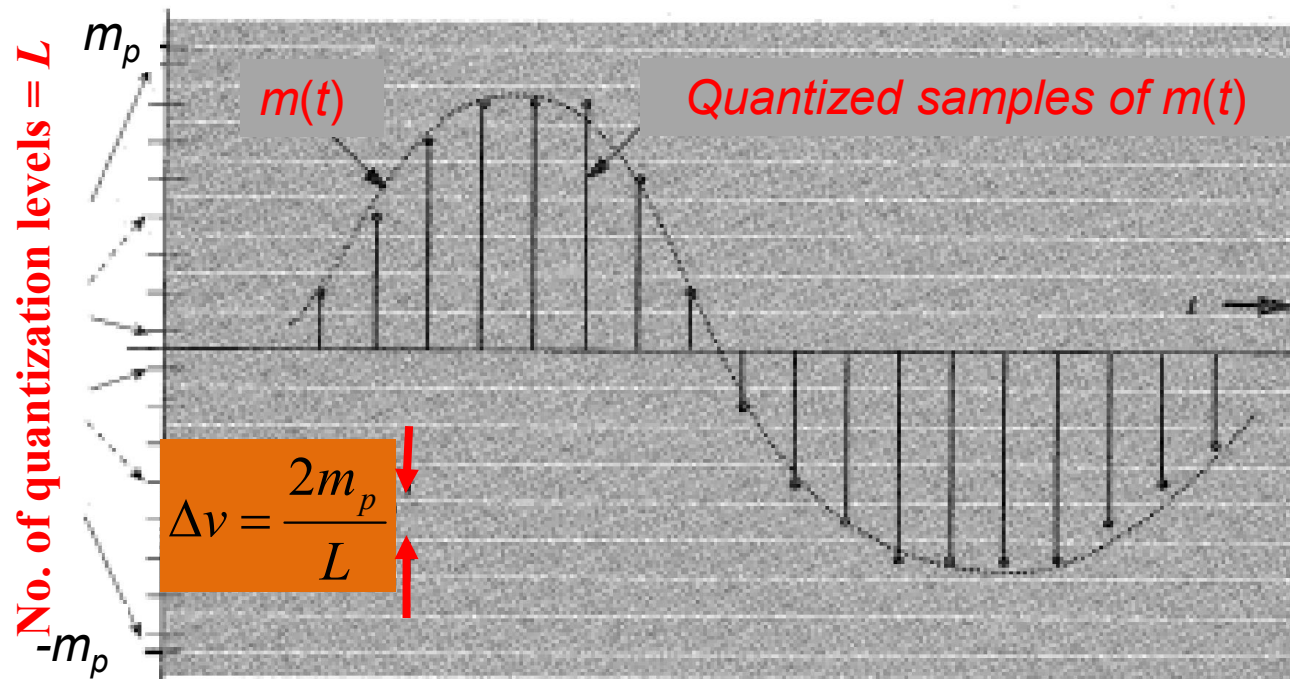
Every sample is represented by $n = \log_2 L$ bits



PCM and Transmission Channel Bandwidth

Causes:

Every sample is represented by $n = \log_2 L$ bits



decreasing L



increasing Δv

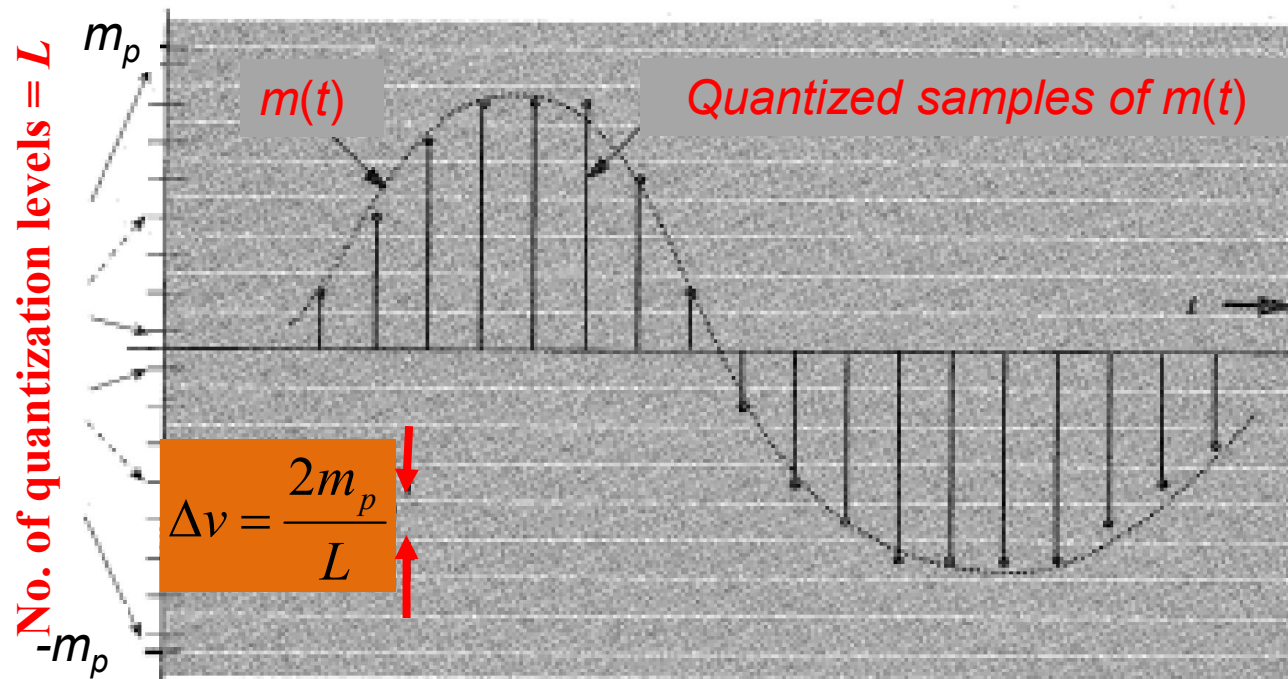


increasing NOISE

PCM and Transmission Channel Bandwidth

Causes:

Every sample is represented by $n = \log_2 L$ bits



decreasing L



increasing Δv



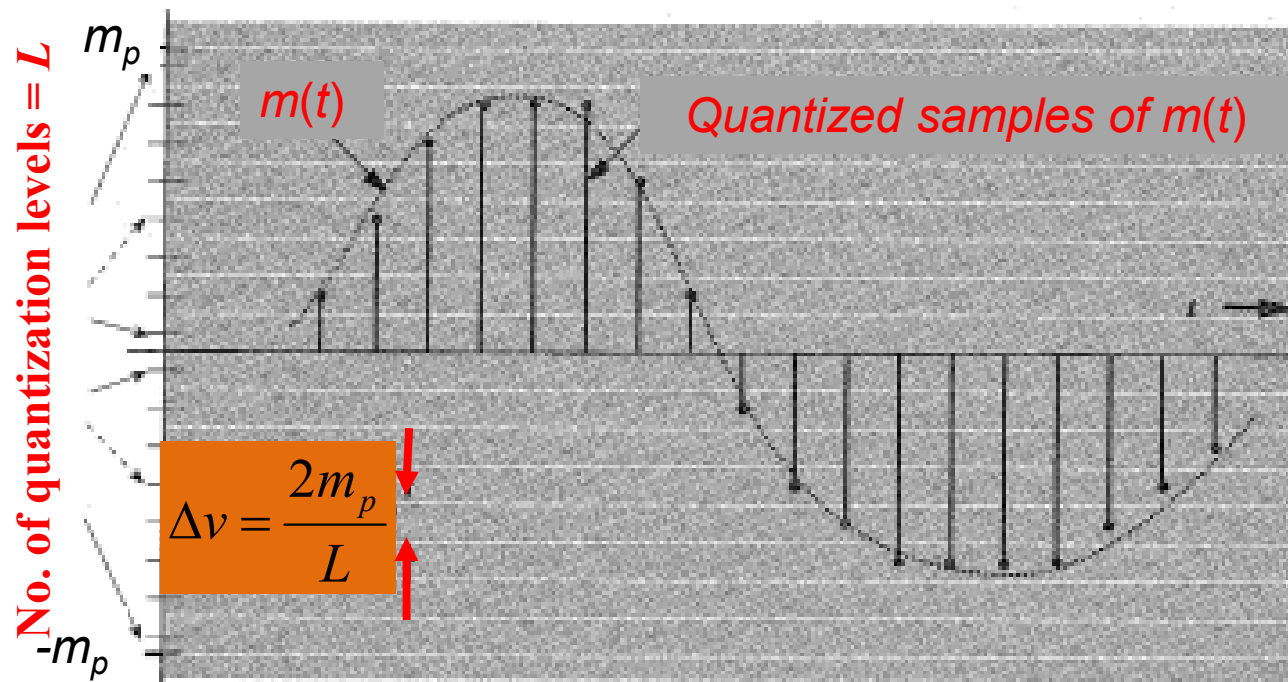
increasing NOISE

Because quantization noise $N_0 = (\Delta v)^2/12$

PCM and Transmission Channel Bandwidth

WAY OUT

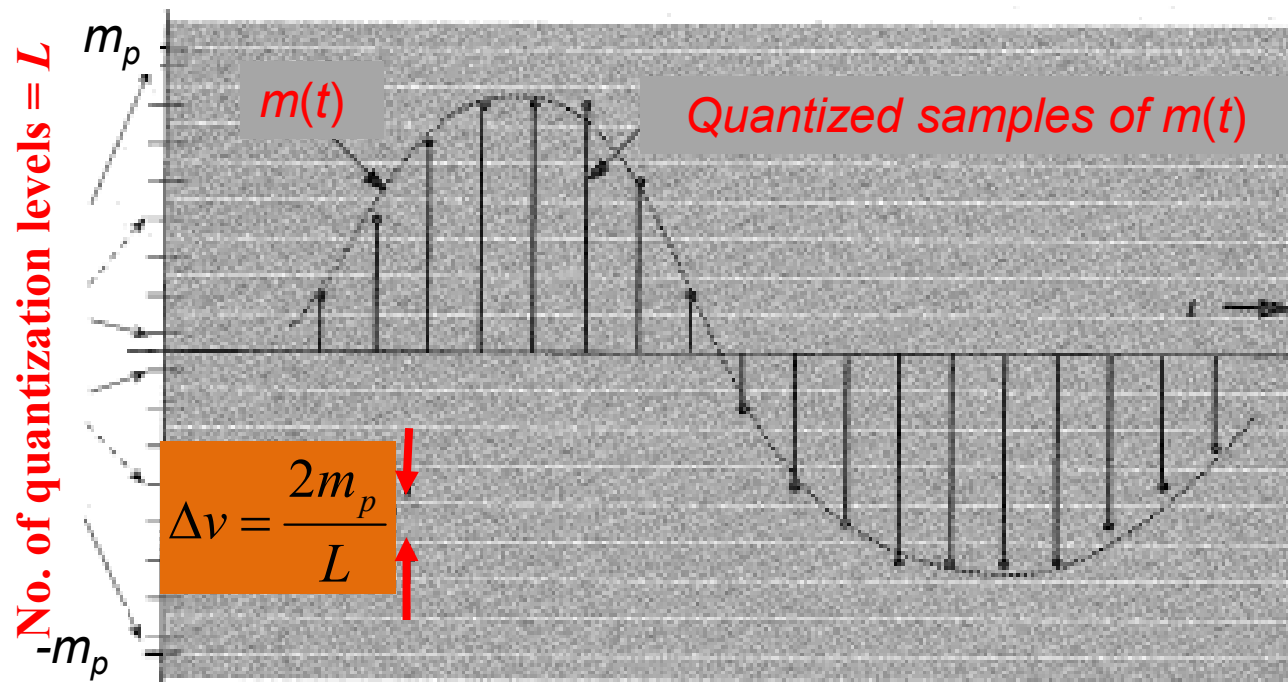
- Successive samples are NOT independent, can be predicted
- Redundancy (CORRELATION) exists among neighboring samples
- Difference can be transmitted instead of actual sample



PCM and Transmission Channel Bandwidth

WAY OUT

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$$d[k] = m[k] - m[k-1]$$

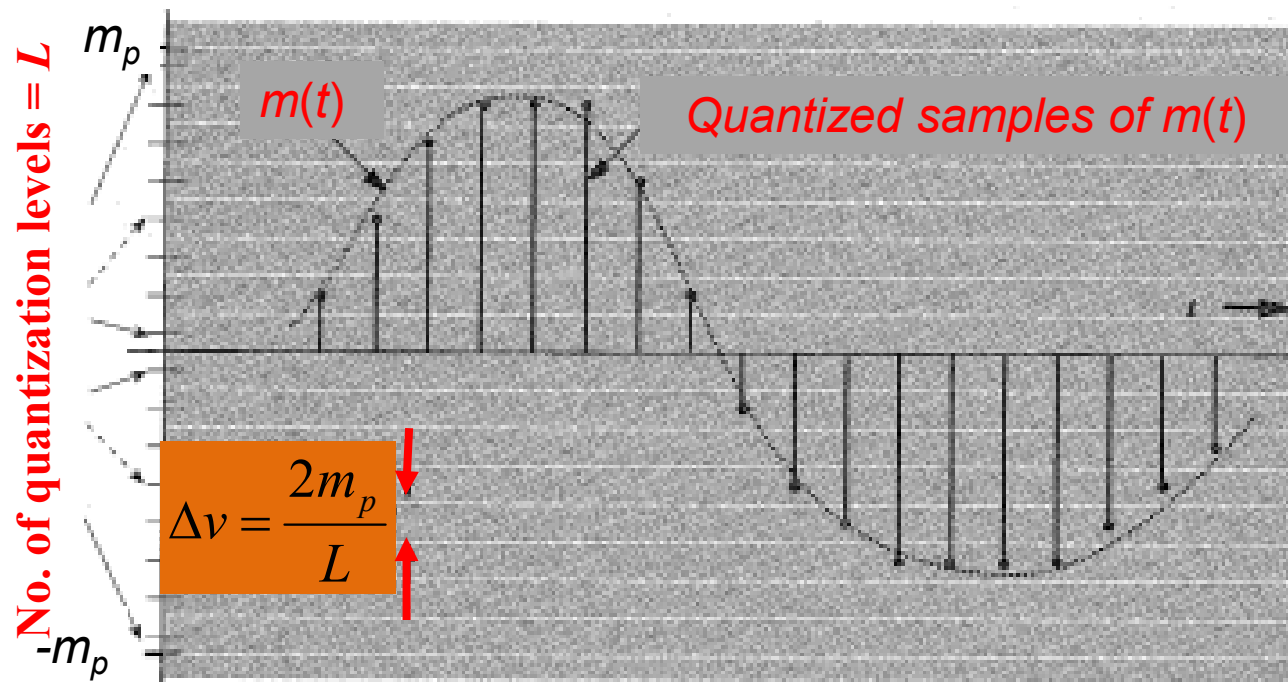
Due to redundancy,

$$d[k] \lll m[k]$$

PCM and Transmission Channel Bandwidth

WAY OUT

- Successive samples are NOT independent, can be predicted
- Redundancy (CORRELATION) exists among neighboring samples
- Difference can be transmitted instead of actual sample



$$d[k] = m[k] - m[k-1]$$



decreases peak d_p



decreases Δv_d

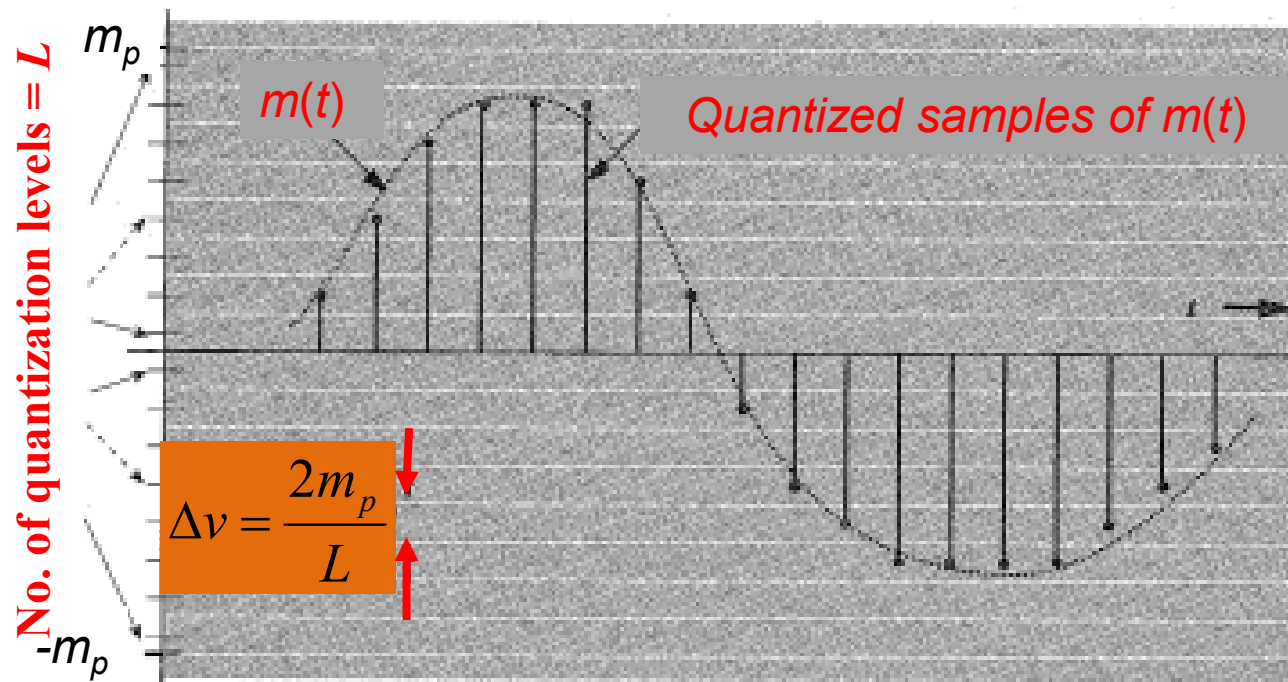


decreases NOISE

PCM and Transmission Channel Bandwidth

Alternatively,

For same NOISE or SNR level, samples can be transmitted by fewer $n = \log_2 L$ bits



$$d[k] = m[k] - m[k-1]$$



decreases peak d_p



decreases Δv_d

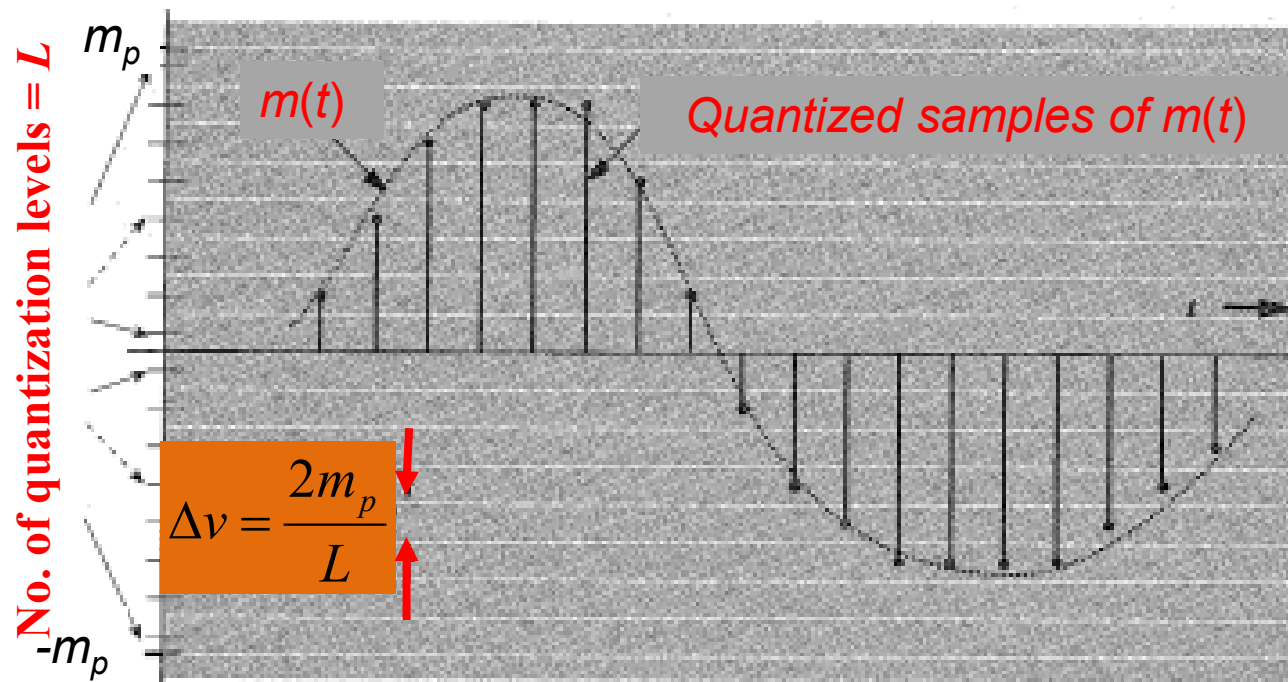


decreases NOISE

PCM and Transmission Channel Bandwidth

$d[k]$ effect

- For same NOISE or SNR, fewer $n = \log_2 L$ bits
- For same n , reduced noise or higher SNR



$$d[k] = m[k] - m[k-1]$$



decreases peak d_p



decreases Δv_d



decreases NOISE

PCM and Transmission Channel Bandwidth

At the receiver

- $m[k]$ is reconstructed from $d[k]$ and previous samples $m[k-1]$

$$m[k] = d[k] + m[k-1]$$

PCM and Transmission Channel Bandwidth

Findings

- $d[k]$ reduces the no. of transmitted bits
- Smaller is $d[k]$, so is n

PCM and Transmission Channel Bandwidth

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Effective prediction of $m[k]$ $\Rightarrow$ further reduction of $d[k]$

PCM and Transmission Channel Bandwidth

Findings

- $d[k]$ reduces the no. of transmitted bits
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Effective prediction of $m[k]$ $\Rightarrow$ further reduction of $d[k]$

Assuming, $\hat{m}[k]$ is the predicted (estimated) value of $m[k]$

✓ Calculate $d[k]$ as $d[k] = m[k] - \hat{m}[k]$

✗ instead of $d[k] = m[k] - m[k-1]$

PCM and Transmission Channel Bandwidth

Better is the prediction, closer is $\hat{m}[k]$ to $m[k]$



Smaller is $d[k]$



Smaller is n

PCM and Transmission Channel Bandwidth

Steps

Predict $\hat{m}[k]$



Calculate the difference, $d[k] = m[k] - \hat{m}[k]$



Represent $d[k]$ in fewer bits and transmit

PCM and Transmission Channel Bandwidth

Differential pulse code modulation (DPCM)

Predict $\hat{m}[k]$



Calculate the difference, $d[k] = m[k] - \hat{m}[k]$



Represent $d[k]$ in fewer bits and transmit

PCM and Transmission Channel Bandwidth

Differential pulse code modulation (DPCM)

difference betn successive samples is a **special case** of DPCM

In DPCM: $d[k] = m[k] - \hat{m}[k]$

In simple difference: $d[k] = m[k] - m[k-1]$

which assumes $\hat{m}[k] = m[k-1]$

Prediction in DPCM

The key step: predicting the future, a kind of *fortune telling*

Predict $\hat{m}[k]$ from past samples $m[k-1]$, $m[k-2]$, $\dots$

Prediction in DPCM

Assuming that, a signal $m(t)$ has derivatives of all orders at t ,

we can predict $m(t+T_s)$ from $m(t)$

where, T_s is sampling interval

Prediction in DPCM

According to Taylor's series,

$$m(t + T_s) = m(t) + T_s \dot{m}(t) + \frac{T_s^2}{2!} \ddot{m}(t) + \frac{T_s^3}{3!} \dddot{m}(t) + \dots$$

Prediction in DPCM

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This formula predicts a future sample $m(t+T_s)$ from past samples of $m(t)$ and its derivatives

Prediction in DPCM

According to Taylor's series,

$$\begin{aligned} m(t + T_s) &= m(t) + T_s \dot{m}(t) + \frac{T_s^2}{2!} \ddot{m}(t) + \frac{T_s^3}{3!} \dddot{m}(t) + \dots \\ &\approx m(t) + T_s \dot{m}(t) \quad \text{for small } T_s \end{aligned}$$

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To replace with discrete time samples,

Assume, $t = kT_s$

k -th sample $m[k] = m(t) = m(kT_s)$

$(k+1)$ -th sample $m[k+1] = m(t+T_s) = m(kT_s+T_s)$

Prediction in DPCM

According to Taylor's series,

$$\begin{aligned} m(t + T_s) &= m(t) + T_s \dot{m}(t) + \frac{T_s^2}{2!} \ddot{m}(t) + \frac{T_s^3}{3!} \dddot{m}(t) + \dots \\ &\approx m(t) + T_s \dot{m}(t) \quad \text{for small } T_s \end{aligned}$$

And, the derivate is

$$\dot{m}(t) = \left[\frac{m[k] - m[k-1]}{T_s} \right]$$

Prediction in DPCM

According to Taylor's series,

$$\begin{aligned} m(t + T_s) &= m(t) + T_s \dot{m}(t) + \frac{T_s^2}{2!} \ddot{m}(t) + \frac{T_s^3}{3!} \dddot{m}(t) + \dots \\ &\approx m(t) + T_s \dot{m}(t) \quad \text{for small } T_s \end{aligned}$$

Finally,

$$\begin{aligned} m[k + 1] &\approx m[k] + T_s \left[\frac{m[k] - m[k - 1]}{T_s} \right] \\ &= 2m[k] - m[k - 1] \end{aligned}$$

Prediction in DPCM

$$m[k+1] \approx 2m[k] - m[k-1]$$

Prediction in DPCM

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$$m[k] \approx 2m[k-1] - m[k-2]$$

Prediction in DPCM

$$m[k+1] \approx 2m[k] - m[k-1]$$



$$m[k] \approx 2m[k-1] - m[k-2]$$

Generalization,

$$m[k] \approx a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$

Prediction in DPCM

Generalized prediction,

$$m[k] \approx a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$

Prediction in DPCM

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The RHS is the predicted value of $m[k]$, that is, $\hat{m}[k]$

Prediction in DPCM

Generalized prediction,

$$m[k] \approx a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$

The RHS is the predicted value of $m[k]$, that is, $\hat{m}[k]$

Therefore,

$$\hat{m}[k] = a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$

Prediction in DPCM

Predicted output,

$$\hat{m}[k] = a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$

Prediction in DPCM

Predicted output,

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The most simplified case:

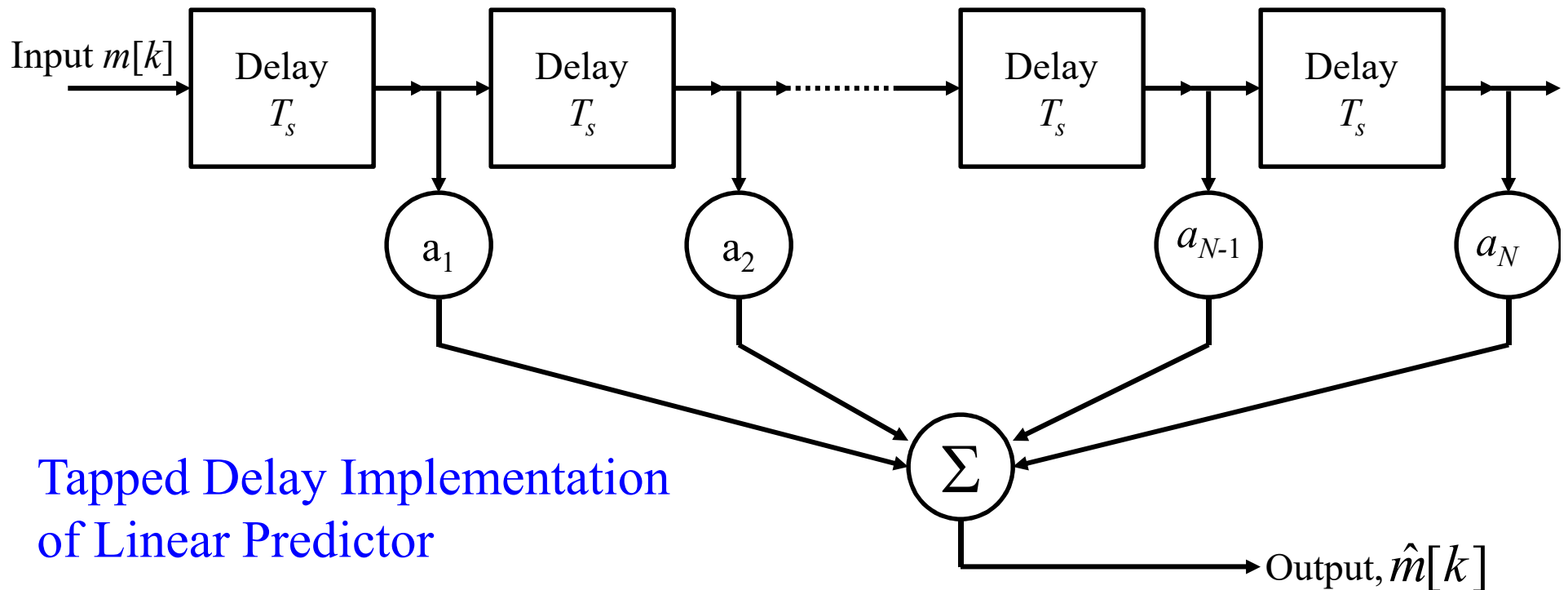
$$\hat{m}[k] = m[k-1]$$

The previous sample is the current sample

Prediction in DPCM

Predicted output,

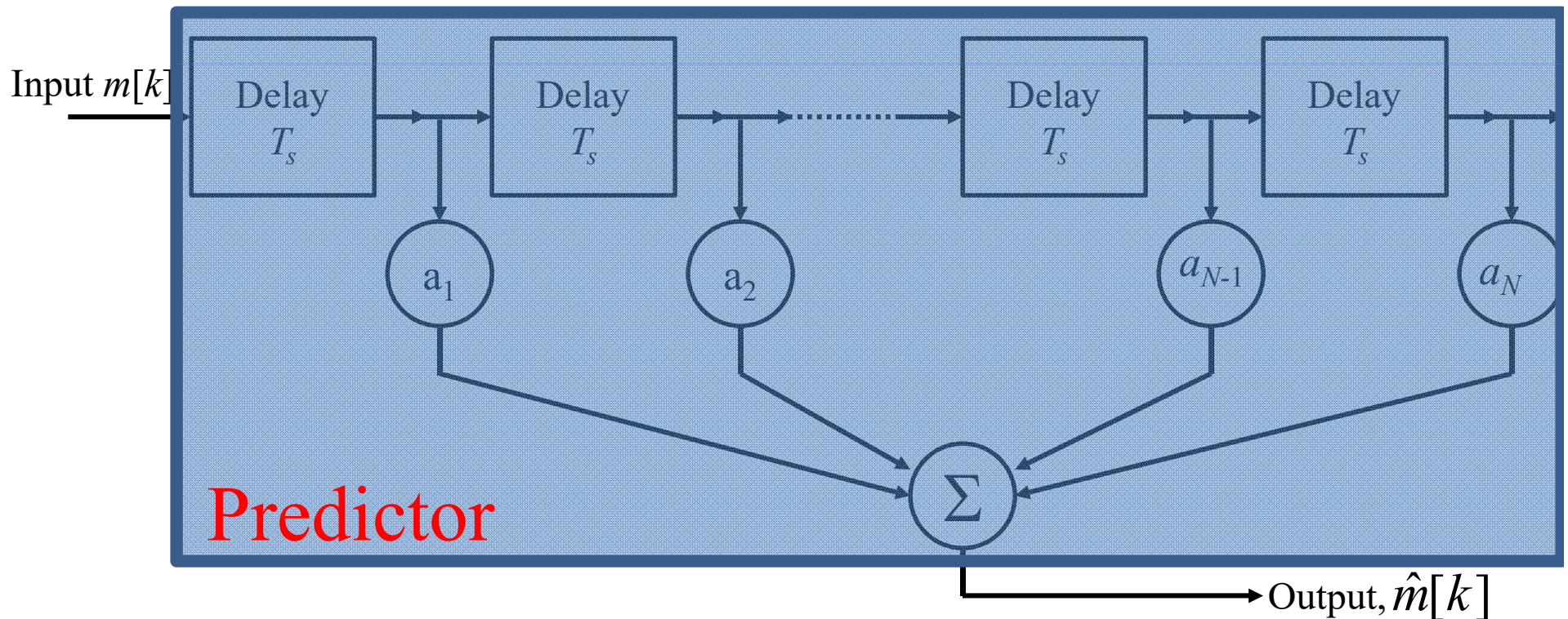
$$\hat{m}[k] = a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$



Prediction in DPCM

Predicted output,

$$\hat{m}[k] = a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$



Prediction in DPCM

Predicted output,

$$\hat{m}[k] = a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N]$$

- We **DO NOT** transmit $m[k]$
- We transmit **difference** betn $m[k]$ and $\hat{m}[k]$

$$d[k] = m[k] - \hat{m}[k]$$

- At receiver, we **recalculate** $m[k]$ from $d[k]$ and $\hat{m}[k]$

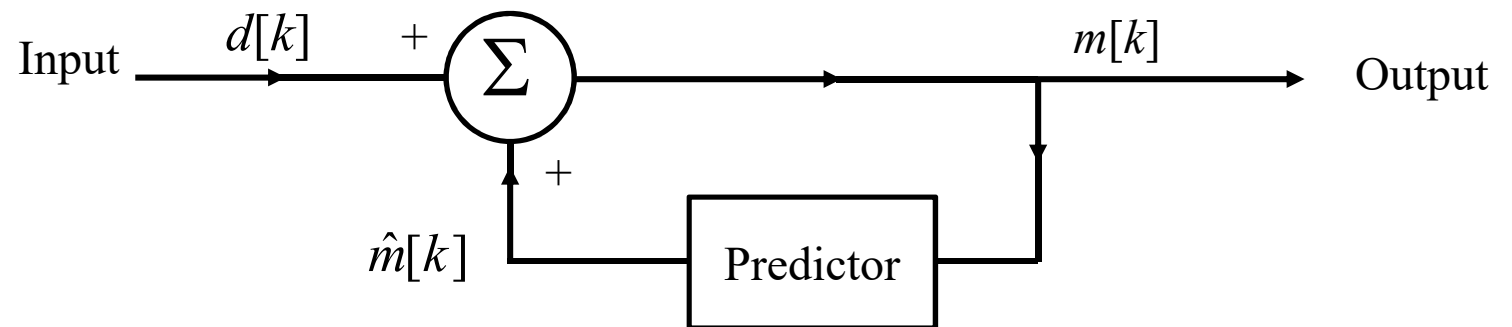
$$m[k] = d[k] + \hat{m}[k]$$

Prediction in DPCM

Transmitter



Receiver



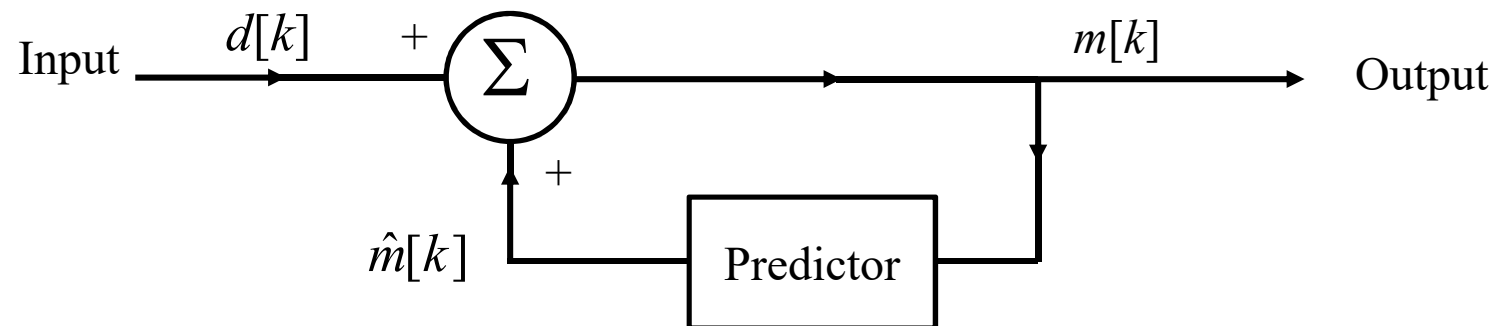
Prediction in DPCM

Transmitter



At Receiver:

$$\begin{aligned} m[k] &= \hat{m}[k] + d[k] \\ &= a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N] + d[k] \end{aligned}$$



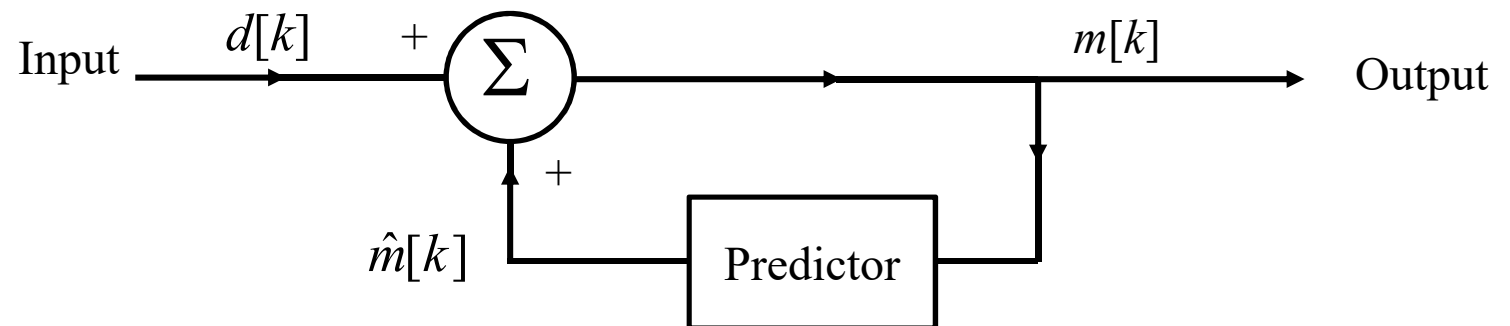
Prediction in DPCM

Transmitter



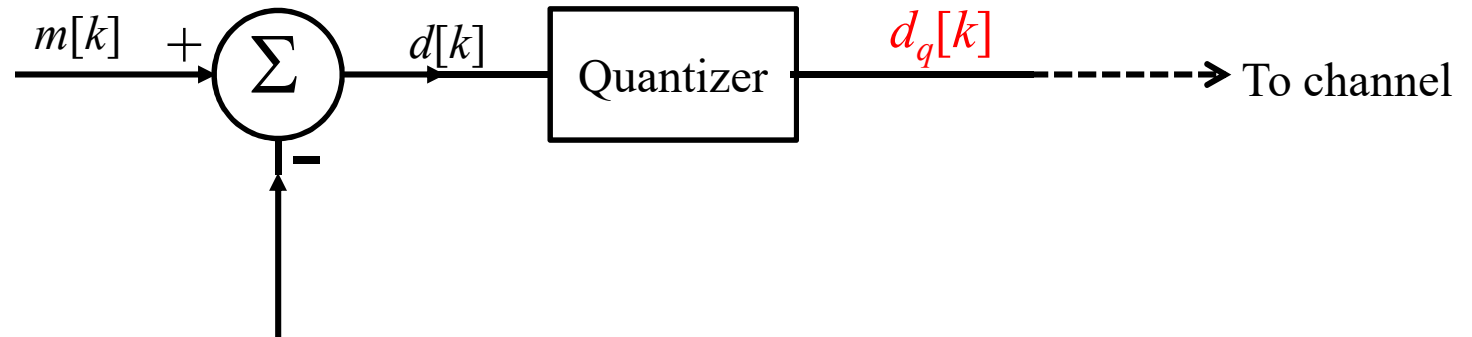
Difficulty at Transmitter:

We **DO NOT** transmit $d[k]$, rather its **quantized** version $d_q[k]$



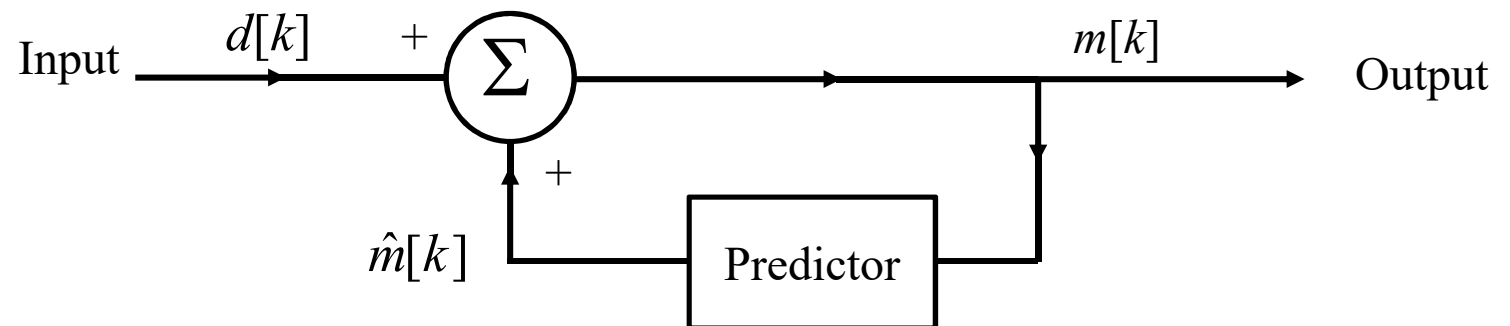
Prediction in DPCM

Transmitter



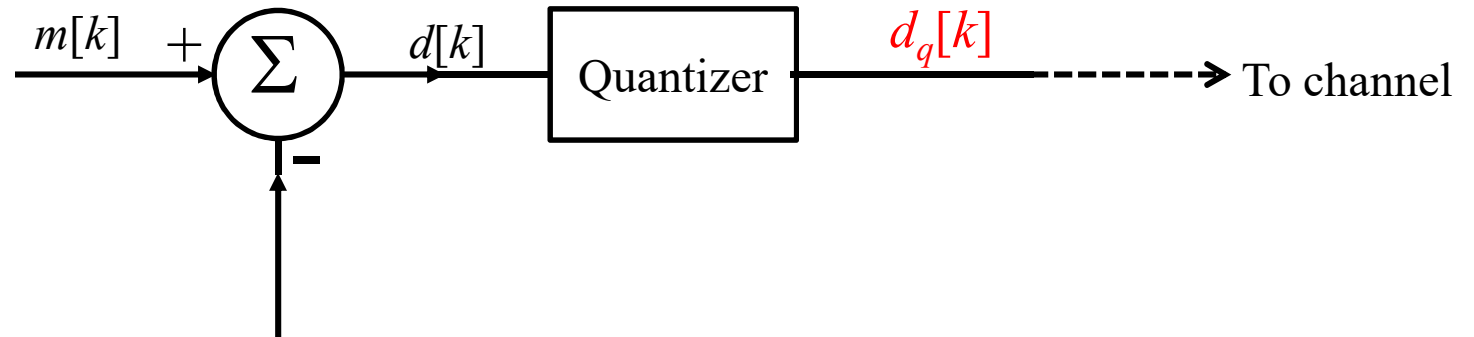
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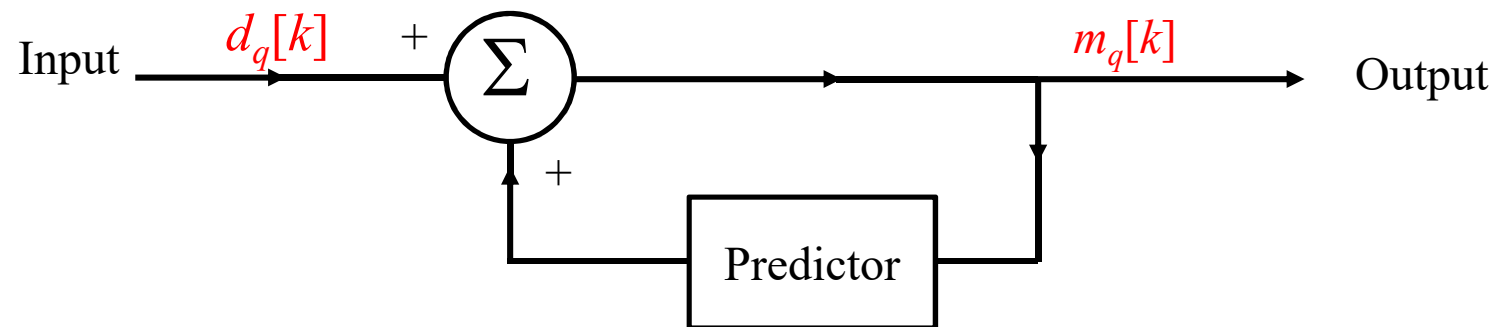


Prediction in DPCM

Transmitter

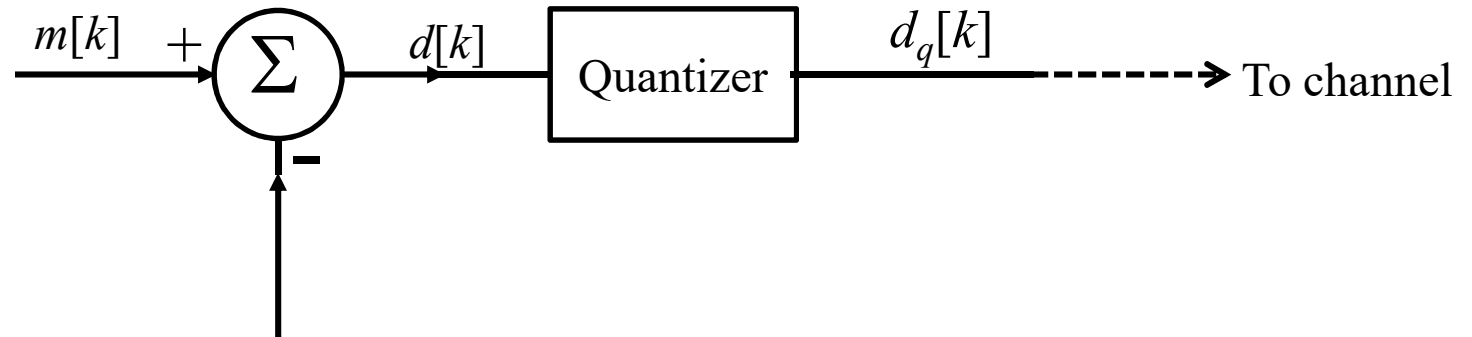


Receiver



Prediction in DPCM

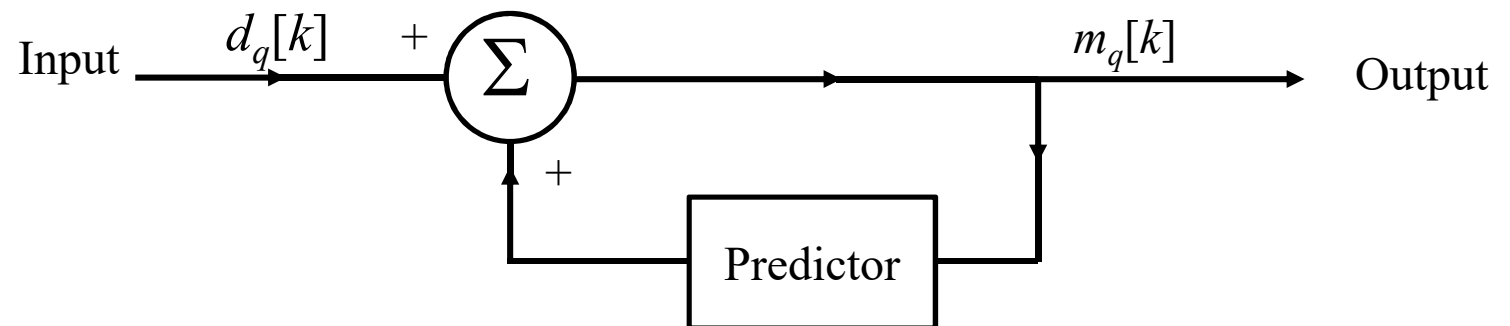
Transmitter



Predictor will generate

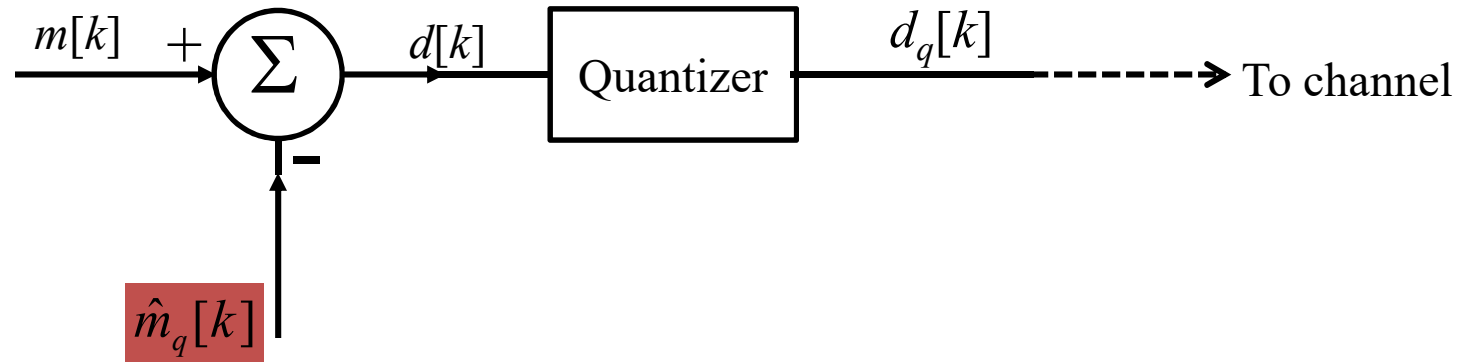
$$\hat{m}_q[k] = a_1 m_q[k-1] + a_2 m_q[k-2] + \dots + a_N m_q[k-N]$$

Receiver



Prediction in DPCM

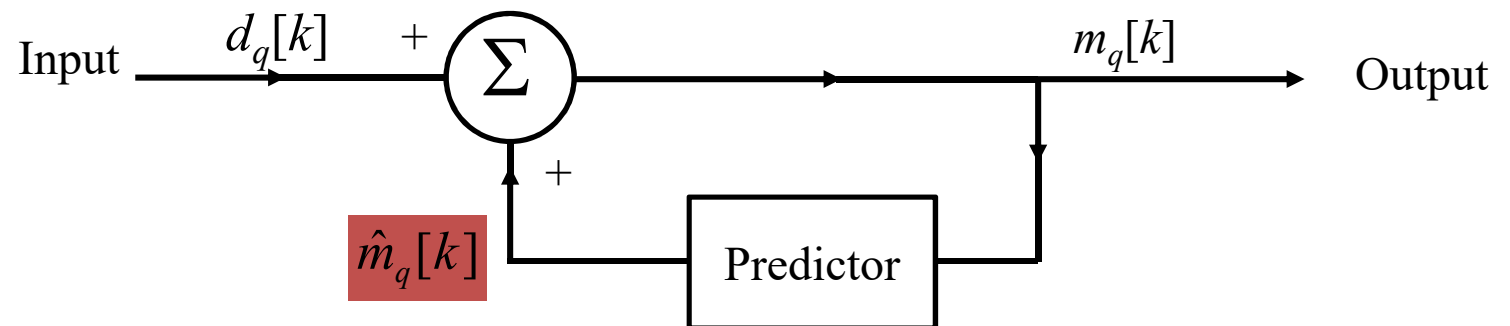
Transmitter



Predictor will generate

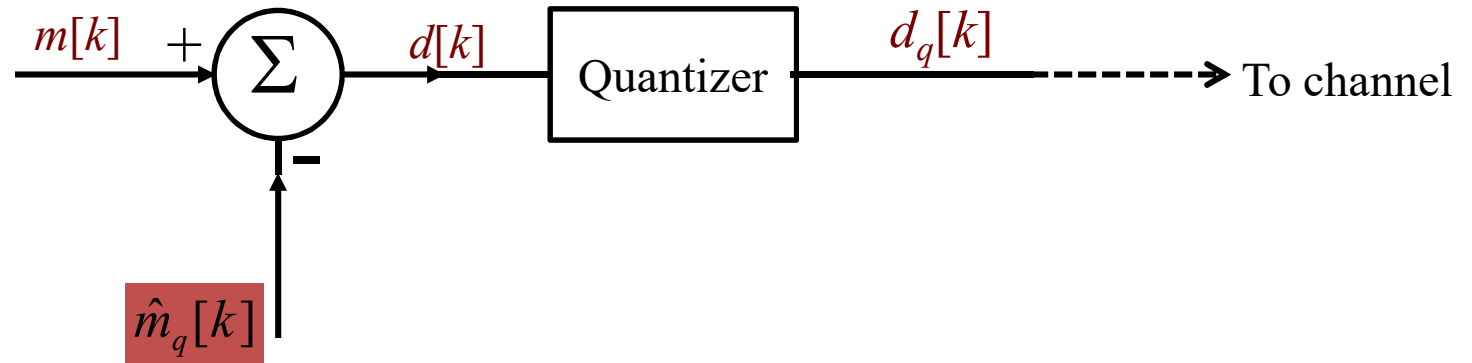
$$\hat{m}_q[k] = a_1 m_q[k-1] + a_2 m_q[k-2] + \dots + a_N m_q[k-N]$$

Receiver



Prediction in DPCM

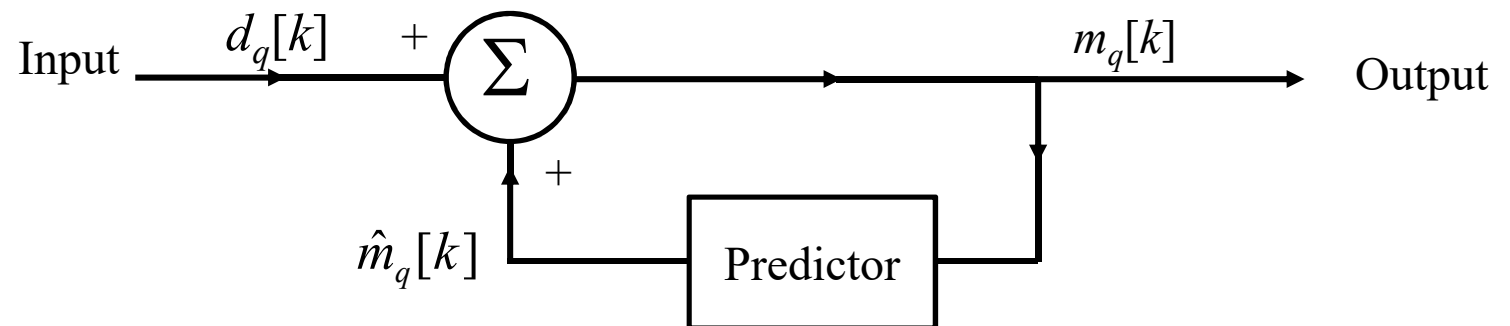
Transmitter



The difference $d[k] = m[k] - \hat{m}_q[k]$ is quantized to yield $d_q[k]$:

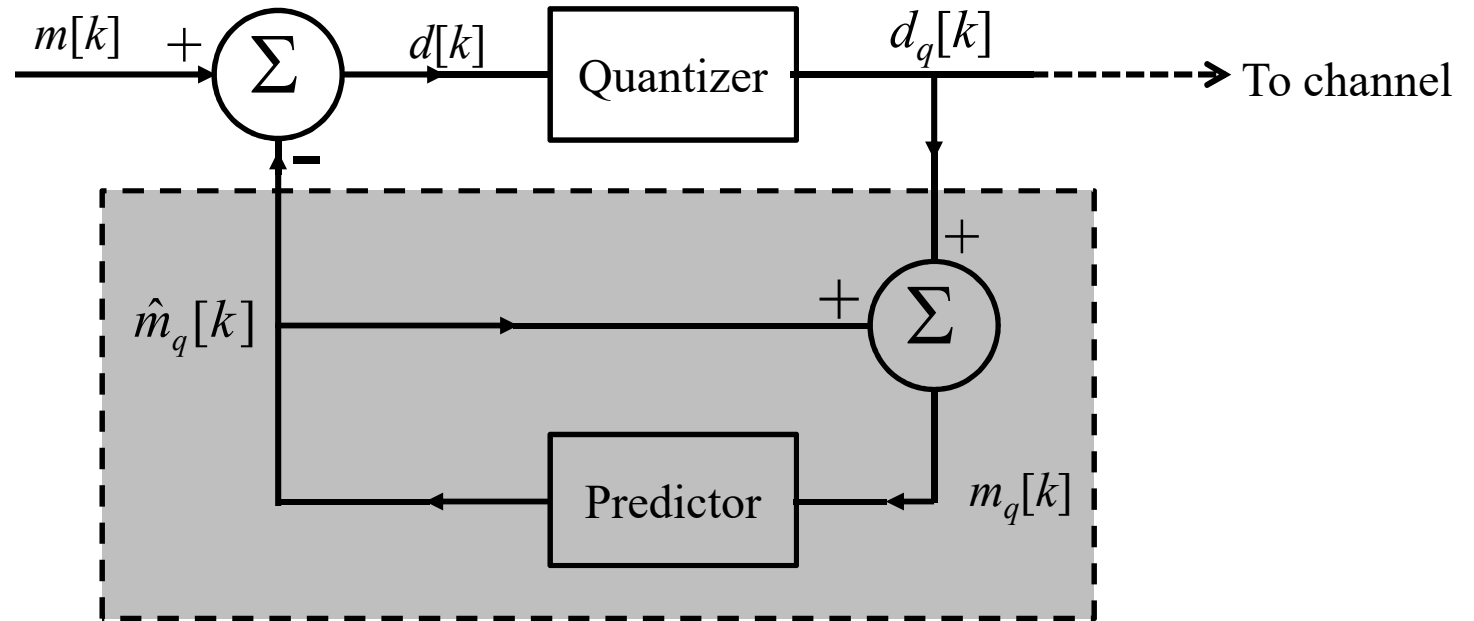
$d_q[k] = d[k] + q[k]$, where $q[k]$ is quantization error

Receiver

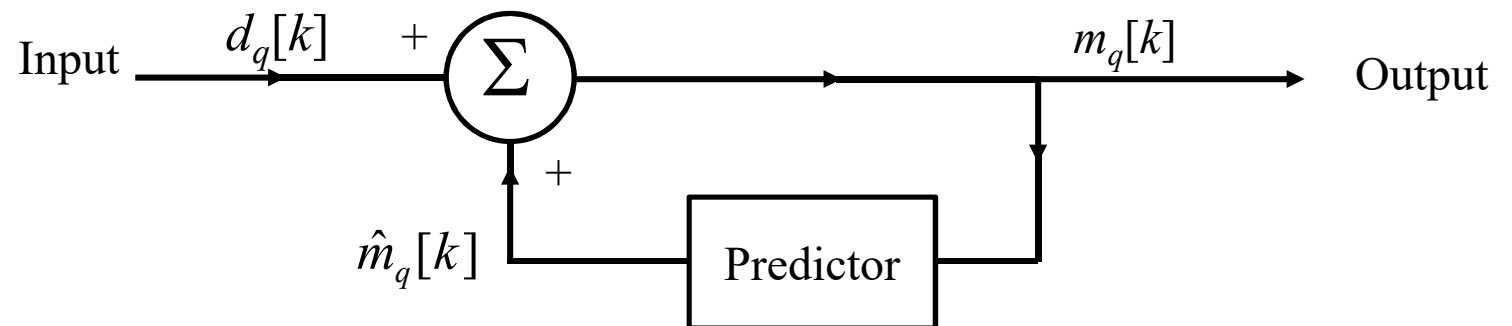


Prediction in DPCM

Transmitter

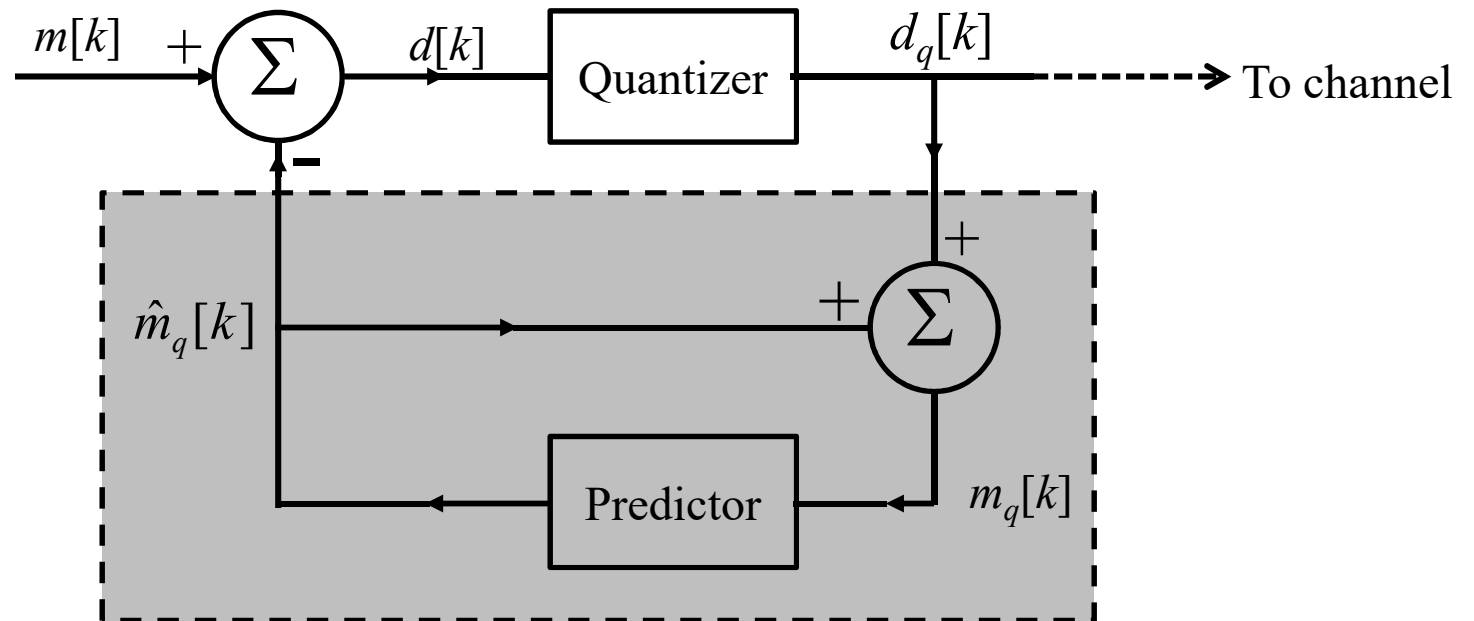


Receiver



Prediction in DPCM

Transmitter

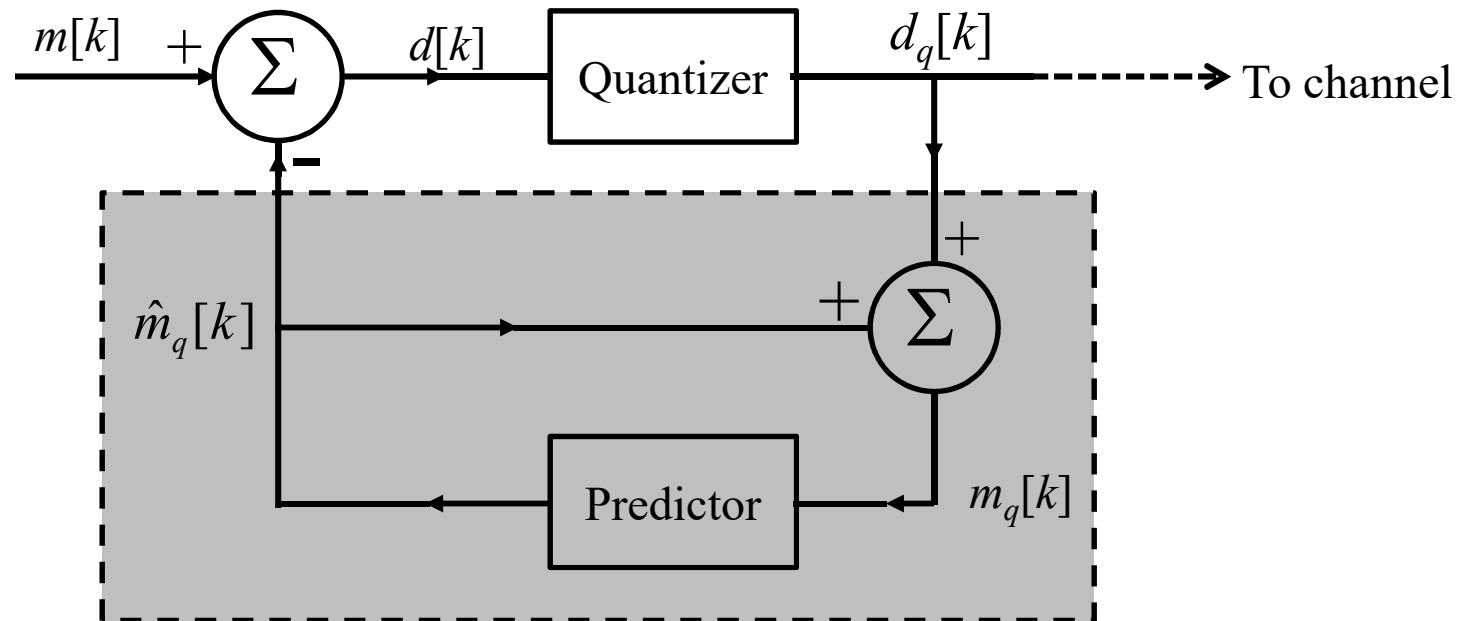


Predictor output $\hat{m}_q[k]$ is feedbacked to its input, that is,

$$m_q[k] = \hat{m}_q[k] + d_q[k]$$

Prediction in DPCM

Transmitter

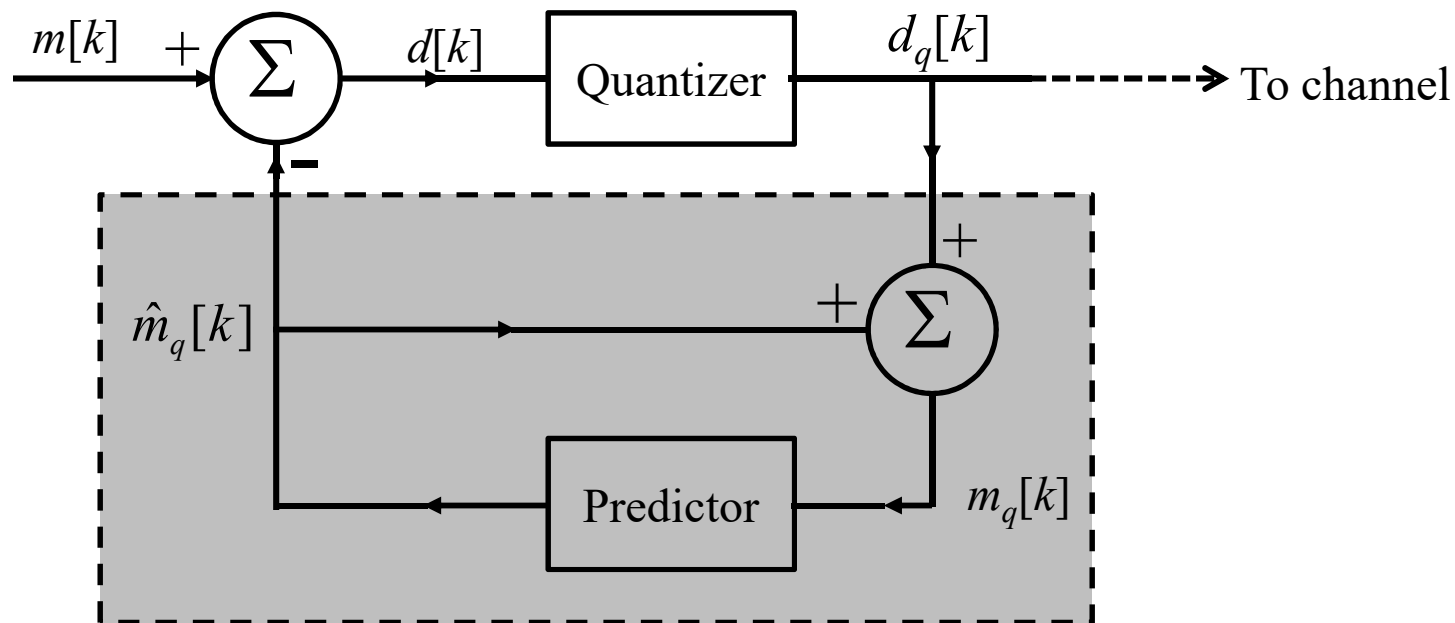


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Prediction in DPCM

Transmitter



Predictor output $\hat{m}_q[k]$ is feedbacked to its input, that is,

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SNR Improvement in DPCM

Let,

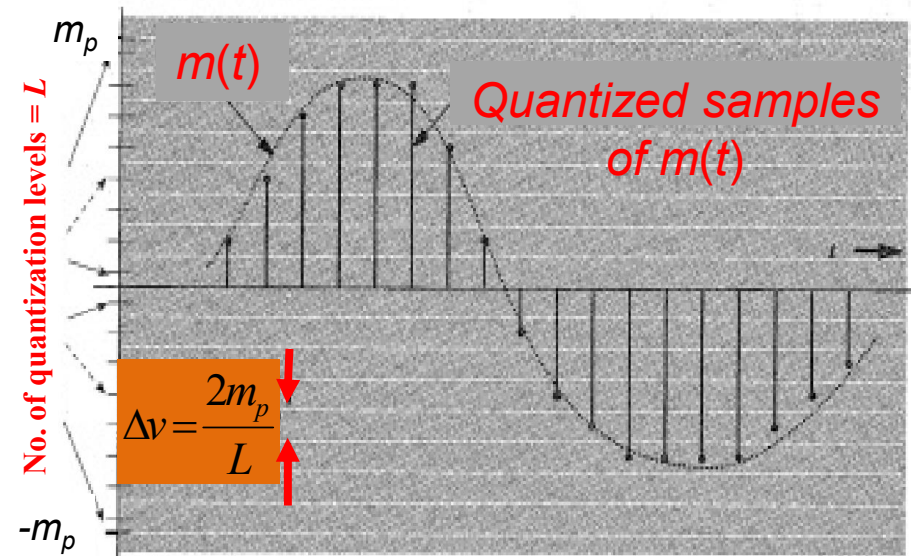
m_p = peak value of message signal $m(t)$

d_p = peak value of difference signal $d(t)$

We know,

$$\Delta v_m = \text{quantization step of } m(t) = \frac{2m_p}{L}$$

$$\Delta v_d = \text{quantization step of } d(t) = \frac{2d_p}{L}$$



Therefore, quantization step **reduction** by a factor of $\frac{d_p}{m_p}$

As quantization noise $N_0 = (\Delta v)^2/12$, it reduces by a factor of $\left(\frac{d_p}{m_p}\right)^2$

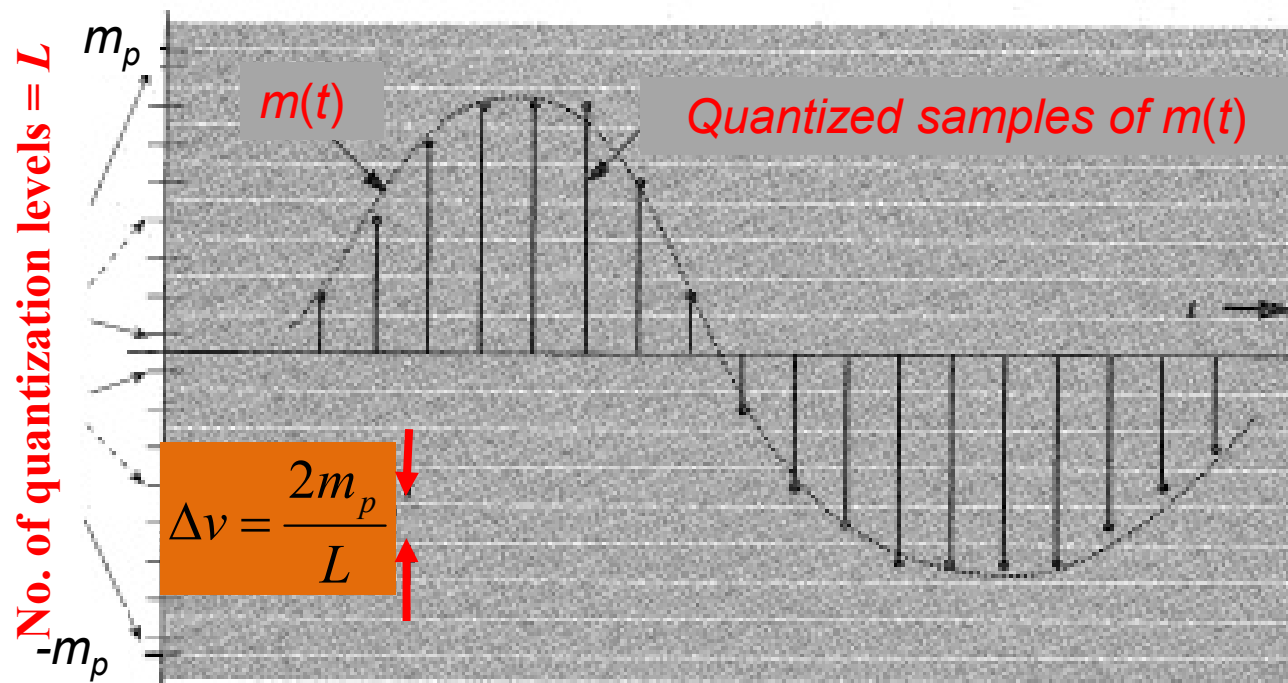
Improvement of DPCM: Delta Modulation

Improvement of DPCM: Delta Modulation

Review

DPCM Motivation:

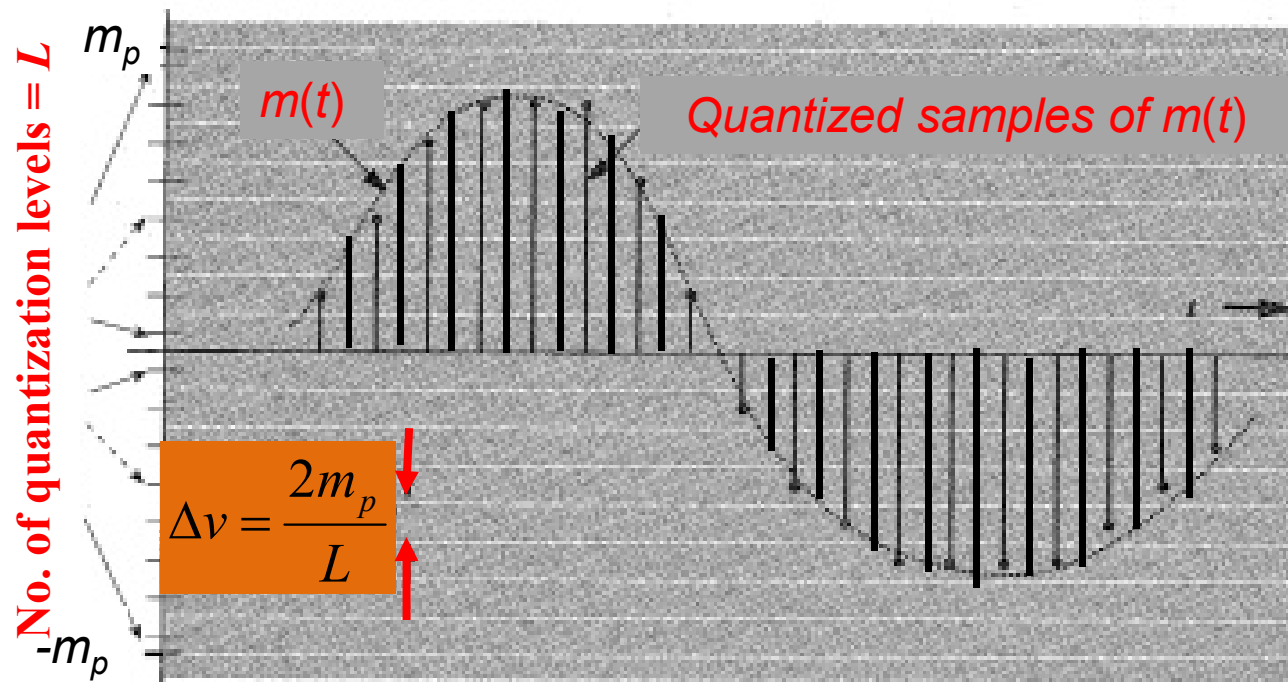
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Improvement of DPCM: Delta Modulation

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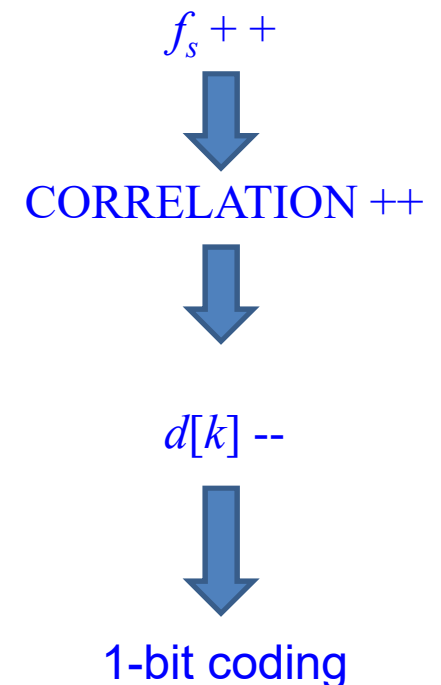
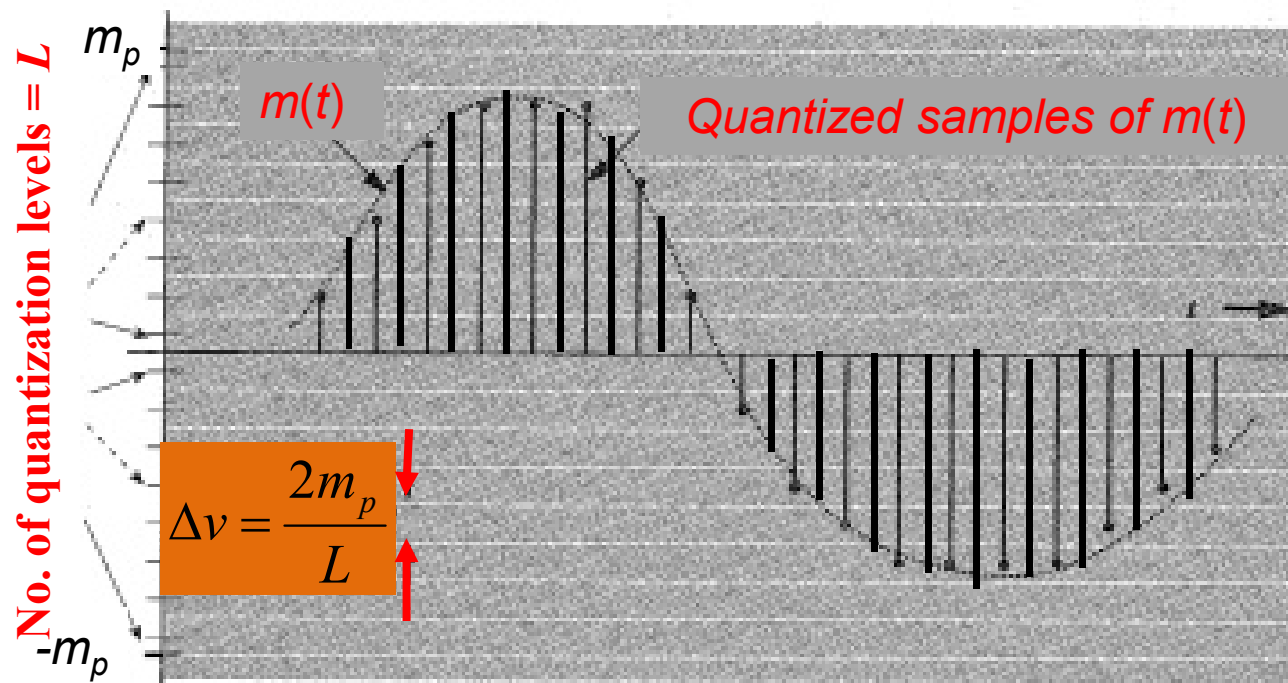


CORRELATION
further increases
if samples are
taken too closely

Improvement of DPCM: Delta Modulation

DPCM Motivation:

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Improvement of DPCM: Delta Modulation

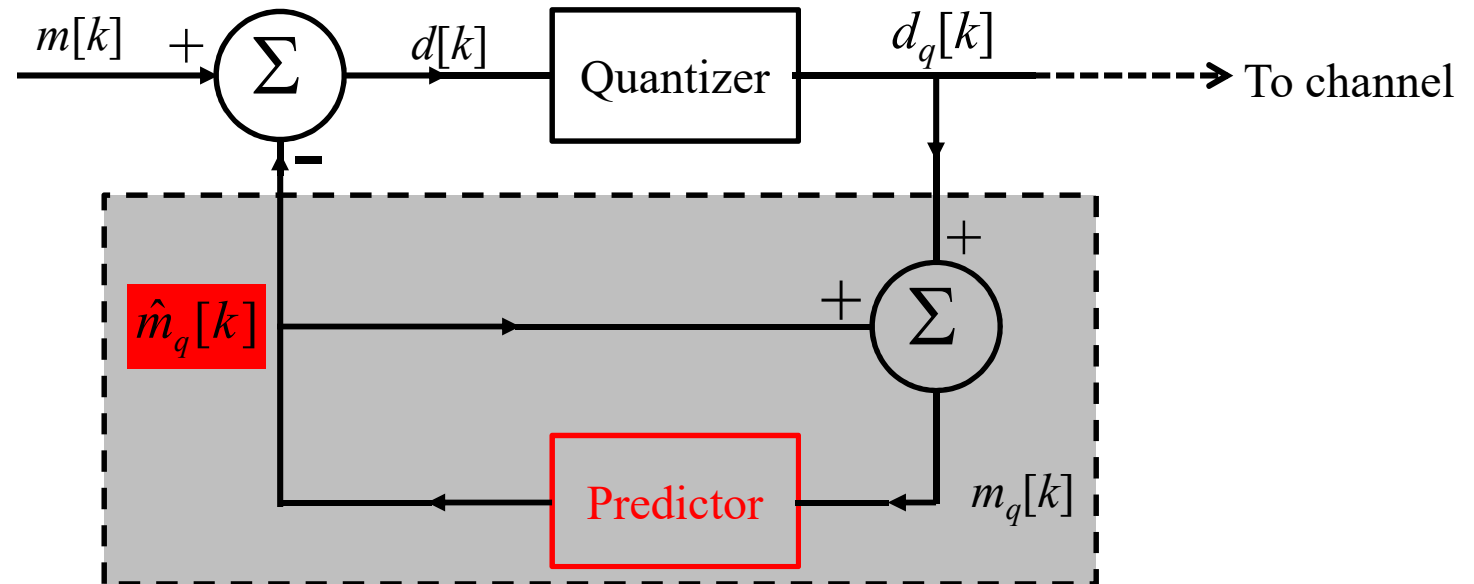
Delta Modulation:

- 1-bit DPCM
- Difference $m[k] - \hat{m}_q[k]$ is quantized in $L = 2$ levels
- Very inexpensive A/D conversion
- $\hat{m}_q[k]$ is assumed to be the previous sample $m_q[k-1]$

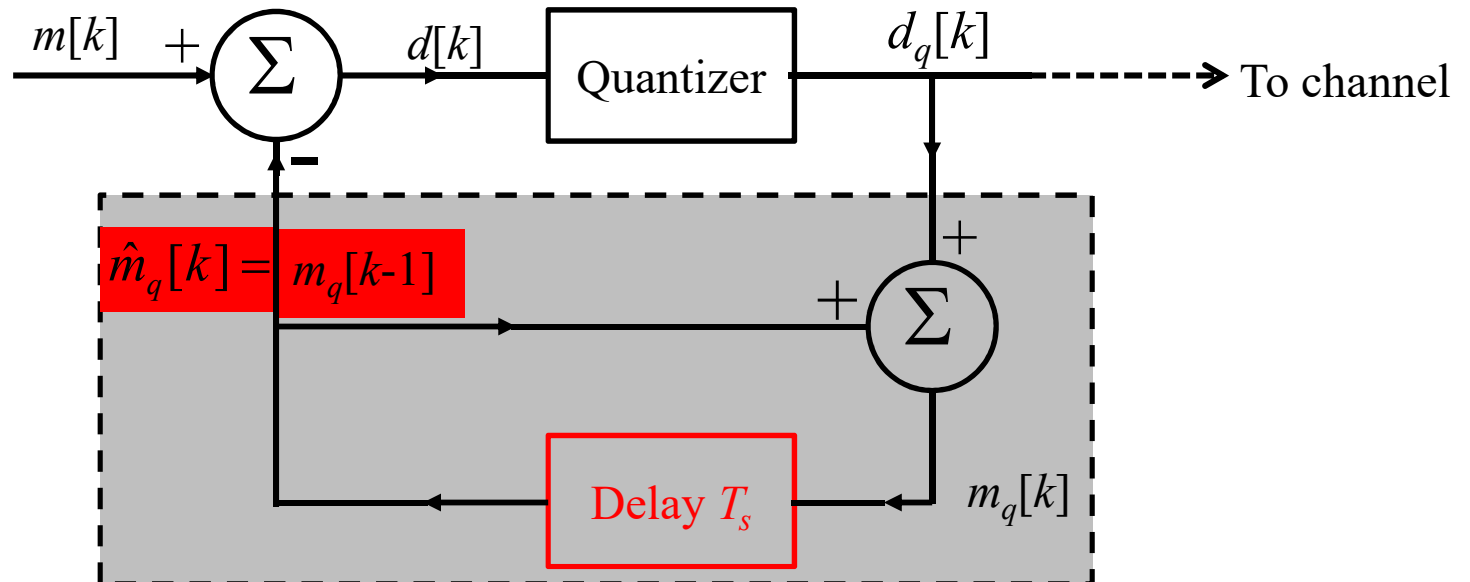
$$\hat{m}_q[k] = m_q[k-1]$$

Transmitters: DPCM Vs. DM

DPCM

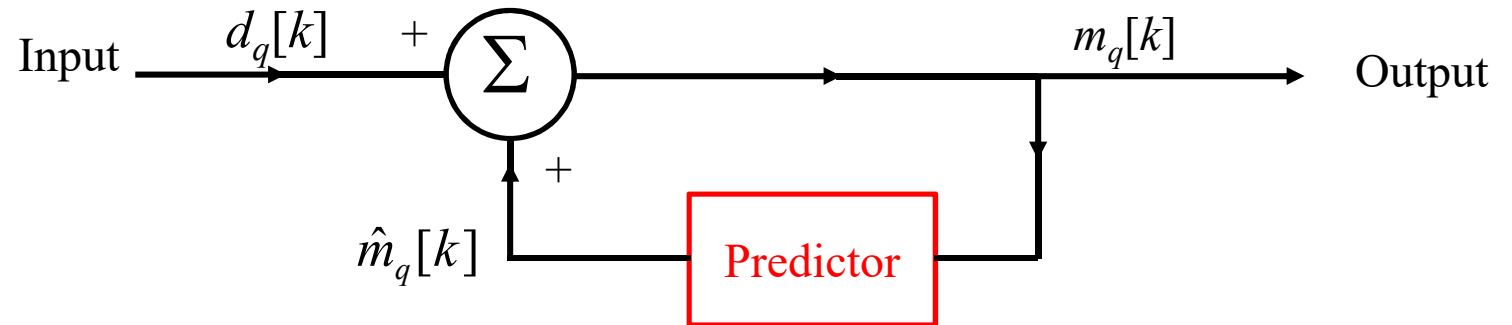


DM

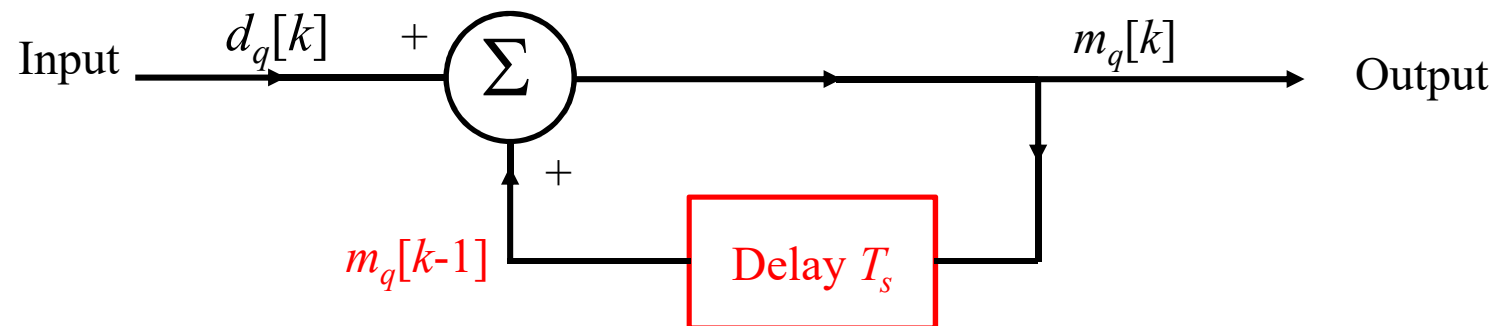


Receivers: DPCM Vs. DM

DPCM

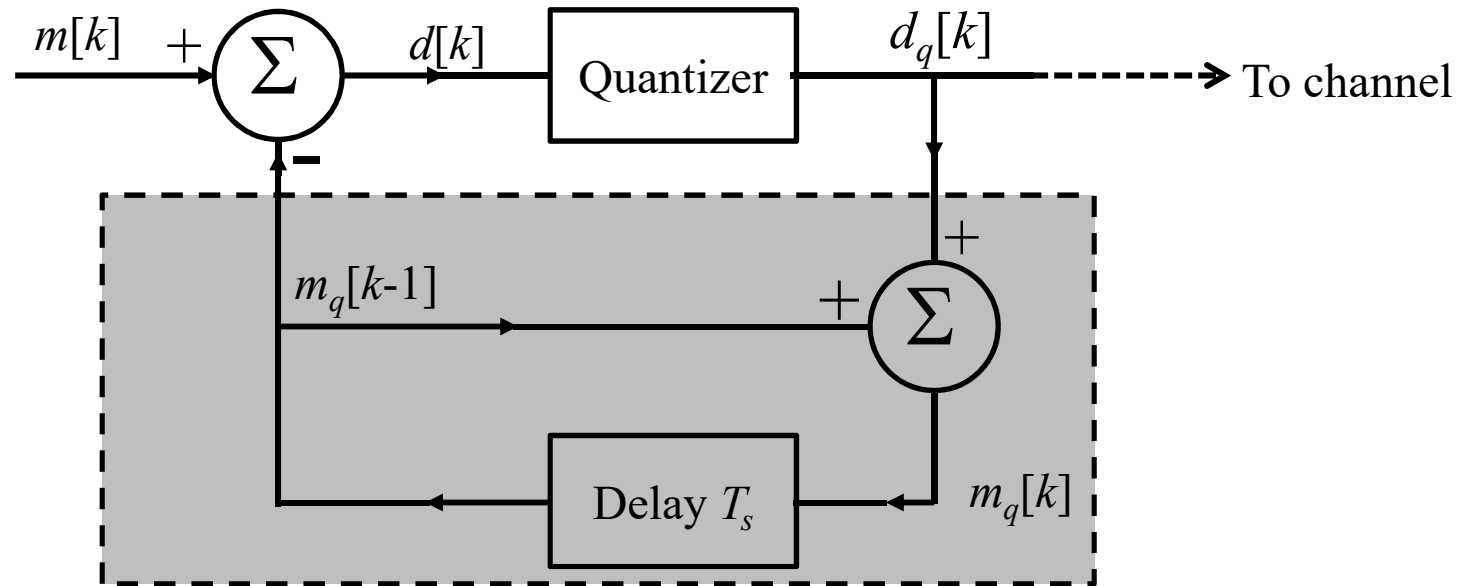


DM

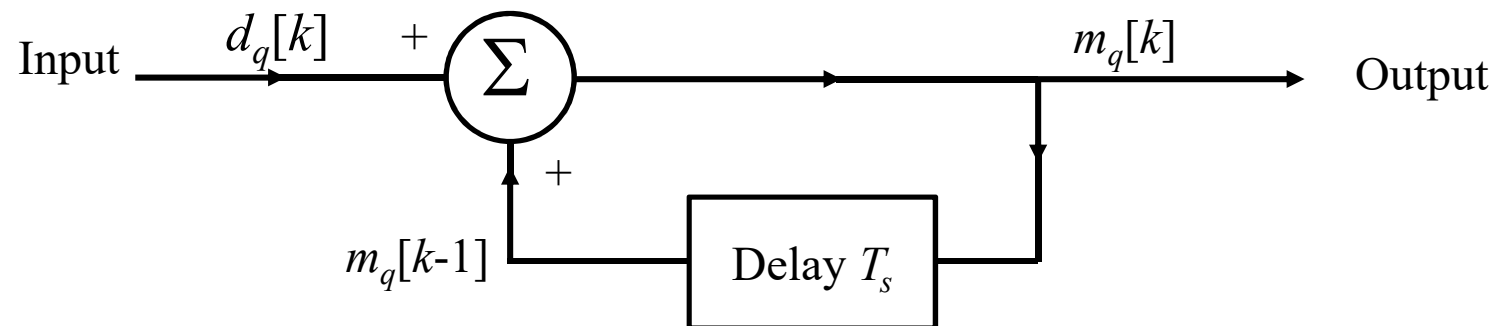


Delta Modulation

Transmitter

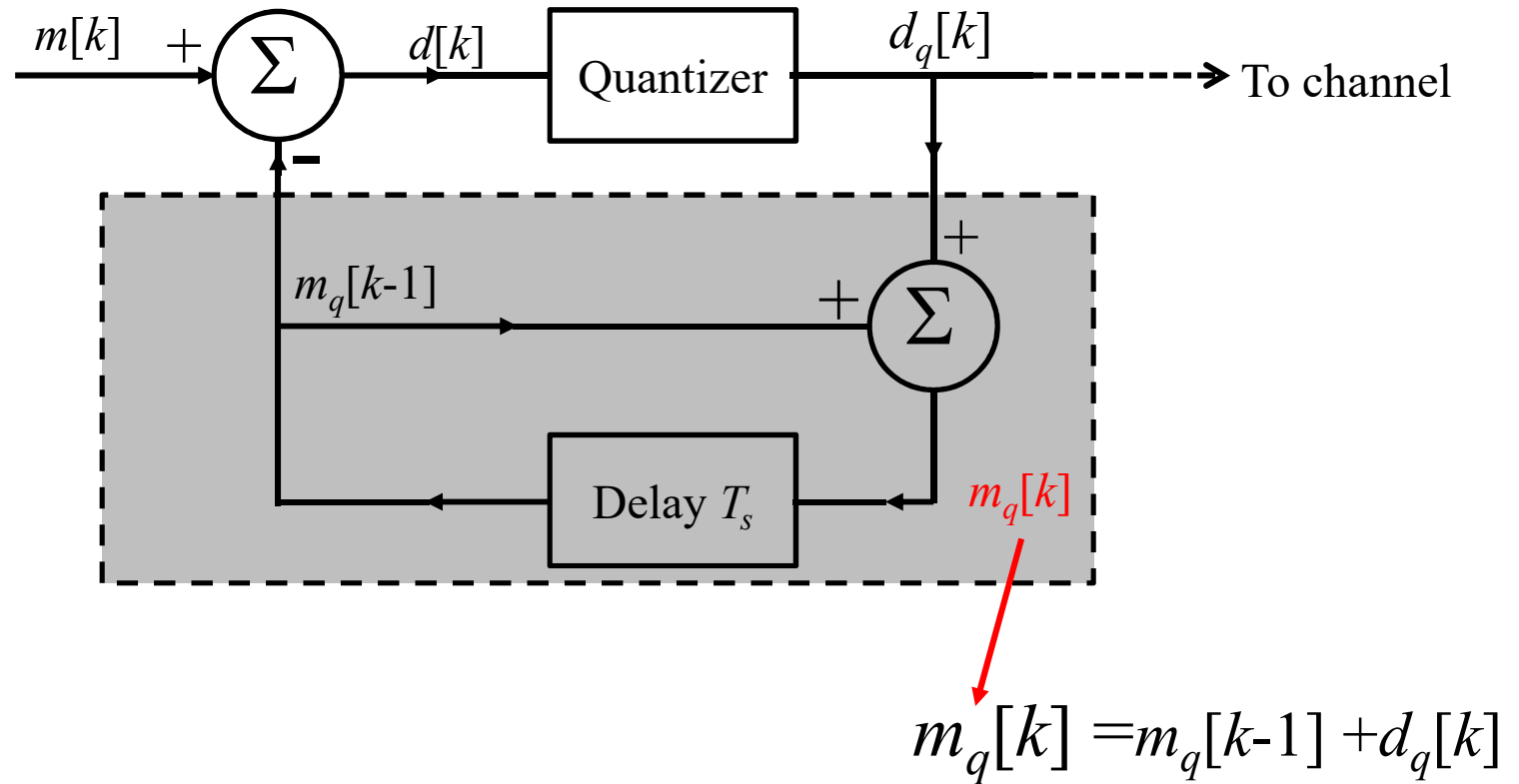


Receiver

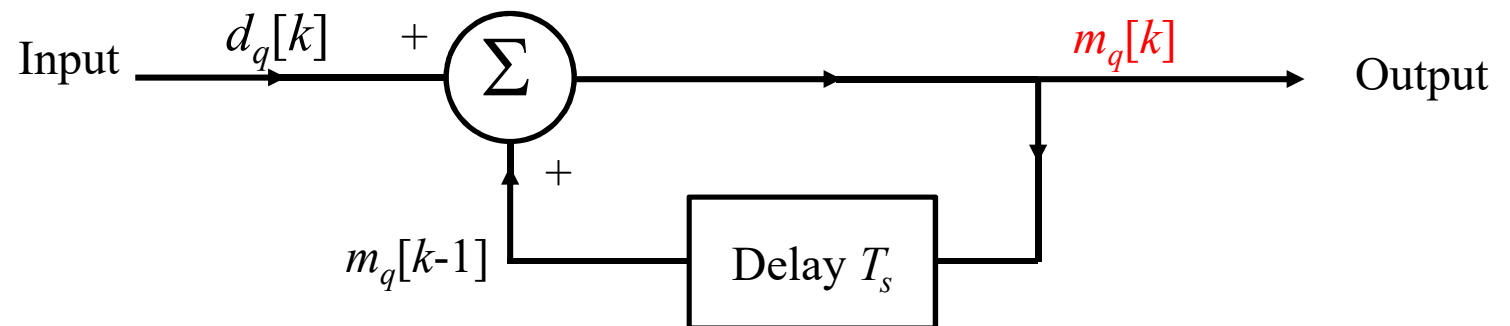


Delta Modulation

Transmitter



Receiver

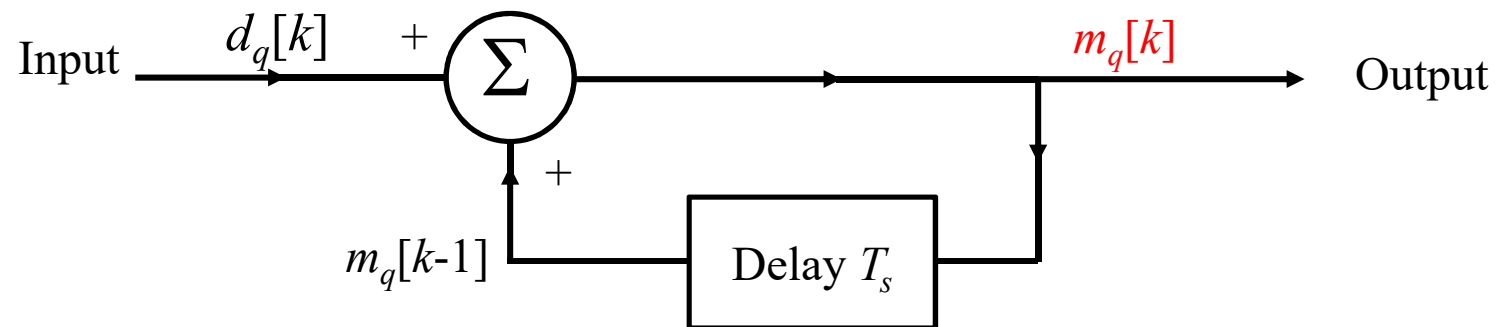


Delta Modulation

$$m_q[k-1] = m_q[k-2] + d_q[k-1]$$

$$m_q[k] = m_q[k-1] + d_q[k]$$

Receiver



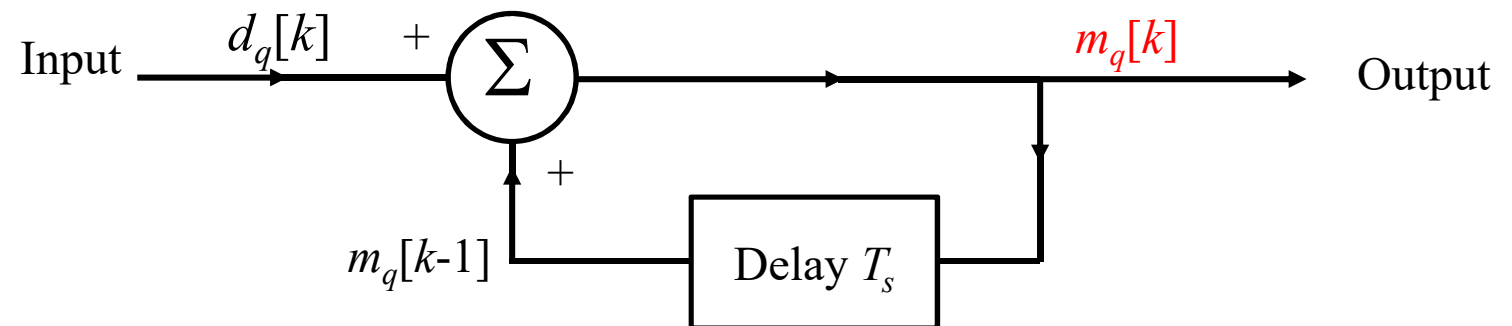
Delta Modulation

$$m_q[k] = m_q[k-2] + d_q[k] + d_q[k-1]$$

$$m_q[k-1] = m_q[k-2] + d_q[k-1]$$

$$m_q[k] = m_q[k-1] + d_q[k]$$

Receiver



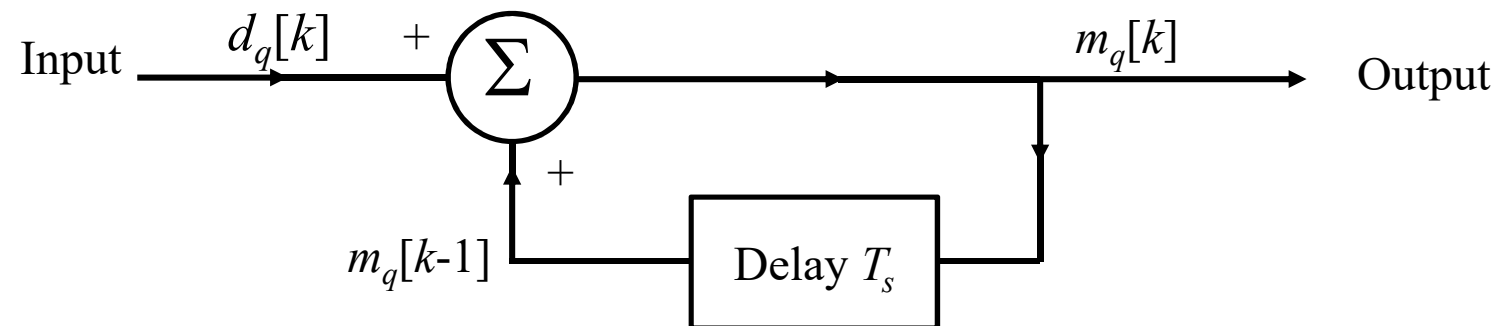
Delta Modulation

$$m_q[k] = m_q[k-2] + d_q[k] + d_q[k-1]$$

Finally, we get

$$m_q[k] = \sum_{m=0}^k d_q[m]$$

Receiver



Delta Modulation

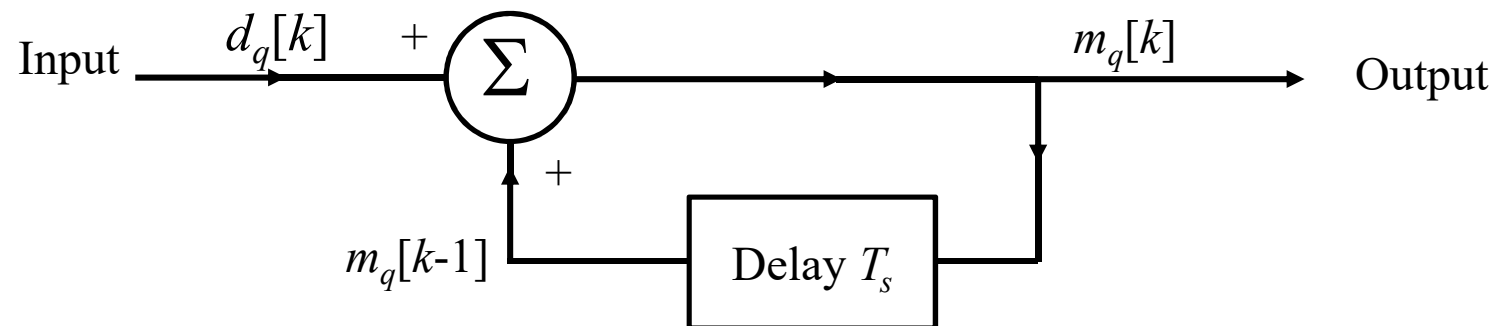
$$m_q[k] = m_q[k-2] + d_q[k] + d_q[k-1]$$

Finally, we get

$$m_q[k] = \sum_{m=0}^k d_q[m]$$

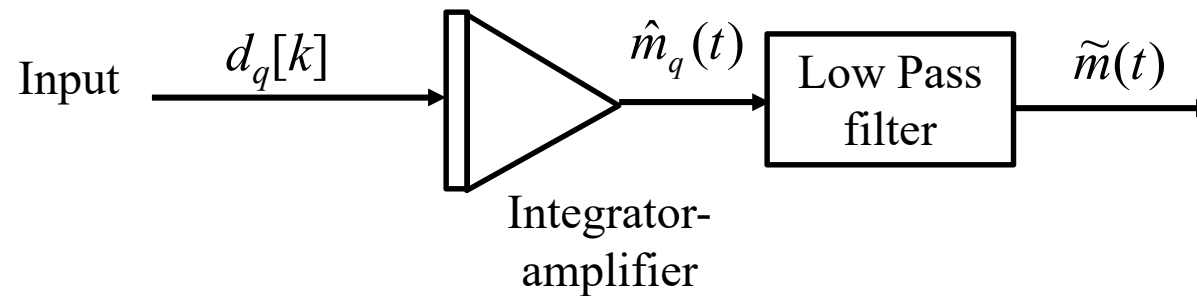
That means, $m_q[k]$ is generated accumulating $d_q[k]$'s

Receiver

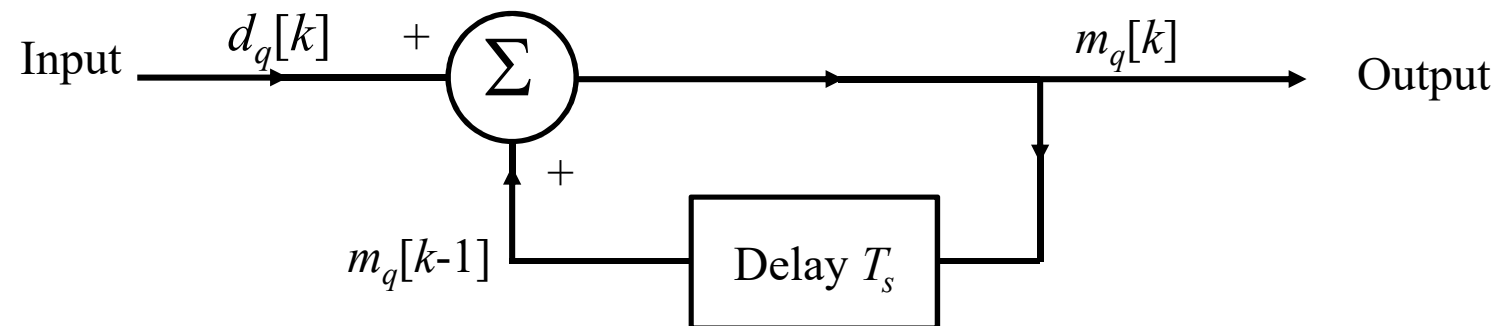


Delta Modulation

Alternate Receiver

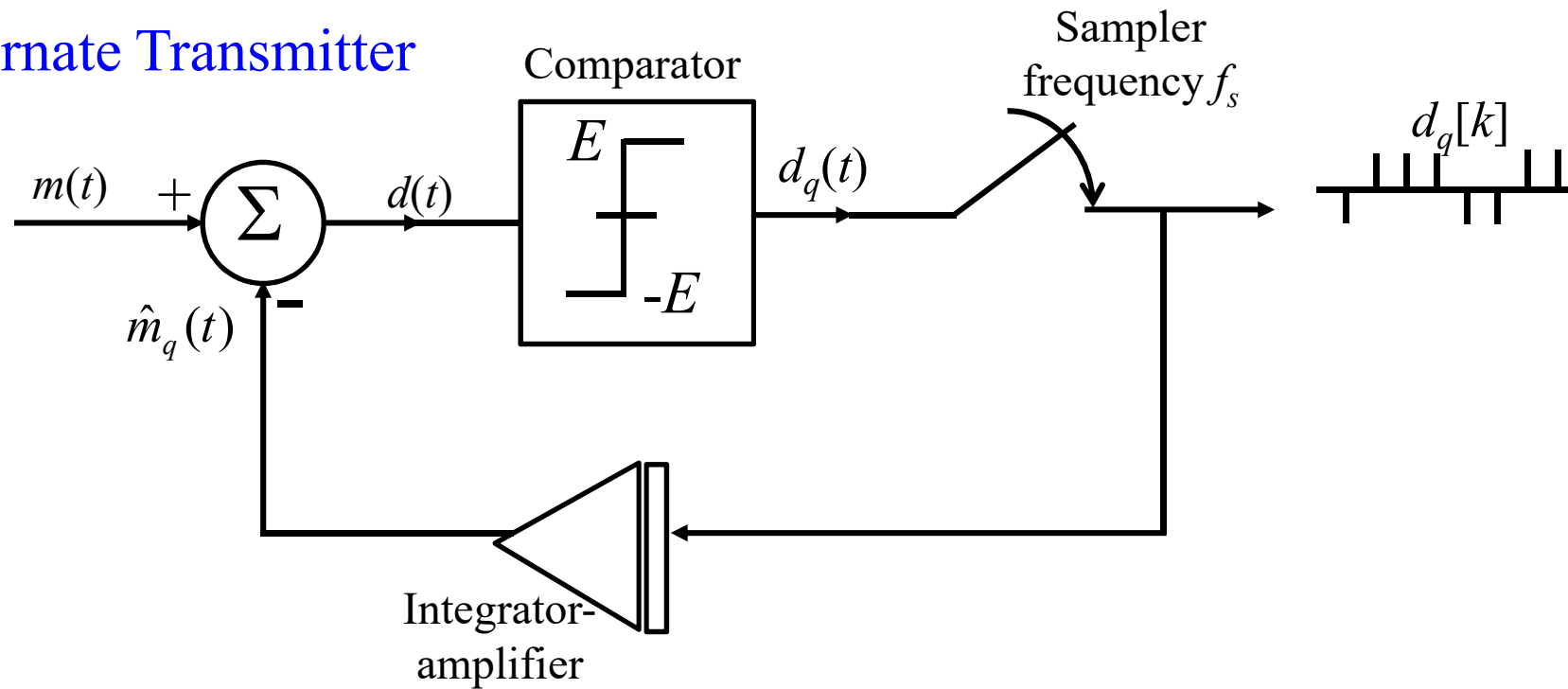


Receiver

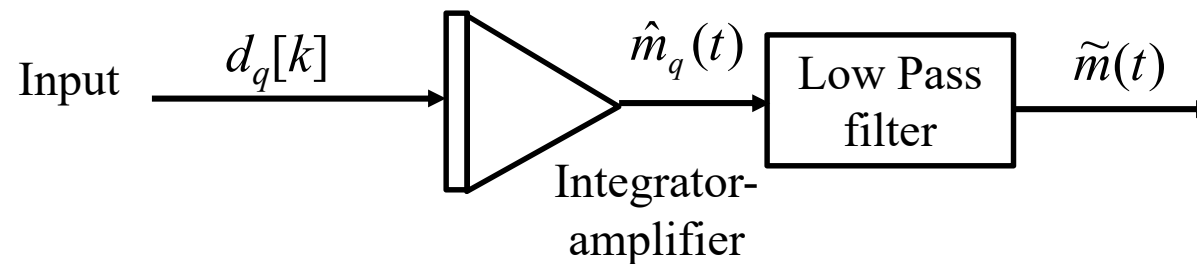


Delta Modulation

Alternate Transmitter

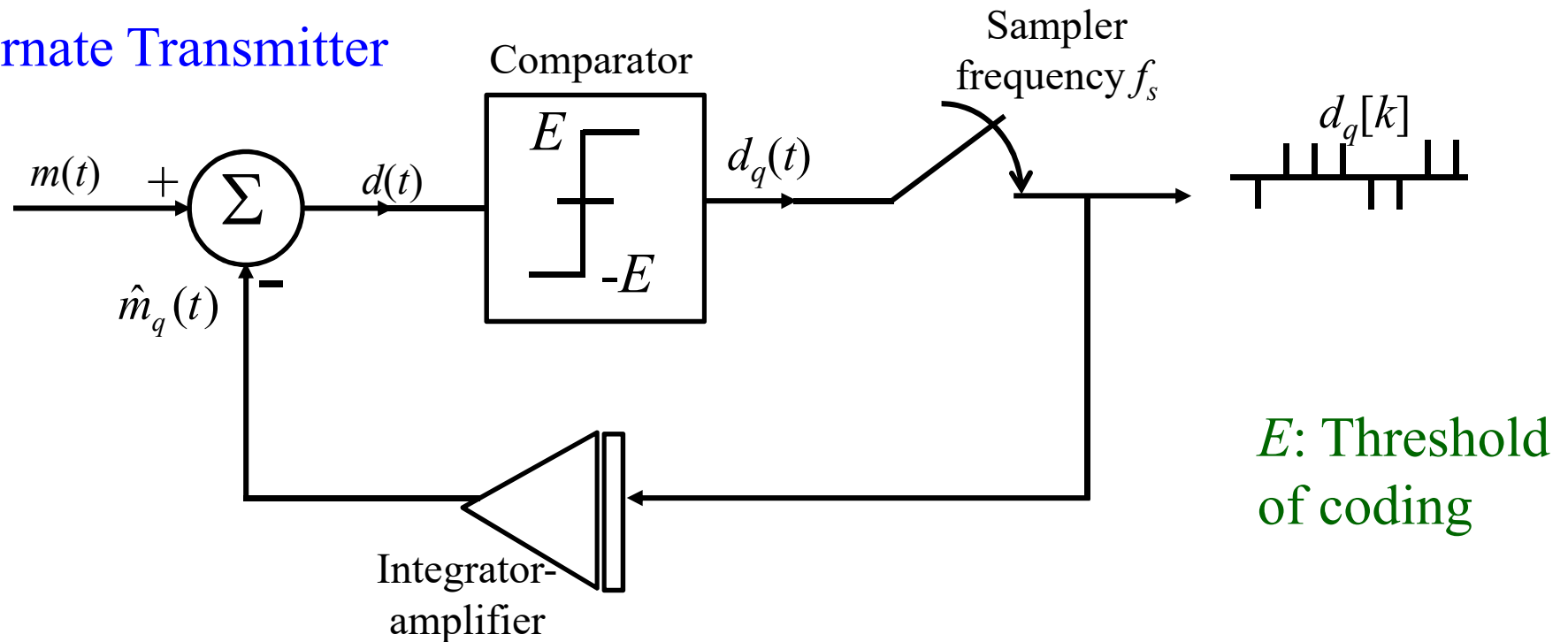


Alternate Receiver



Delta Modulation

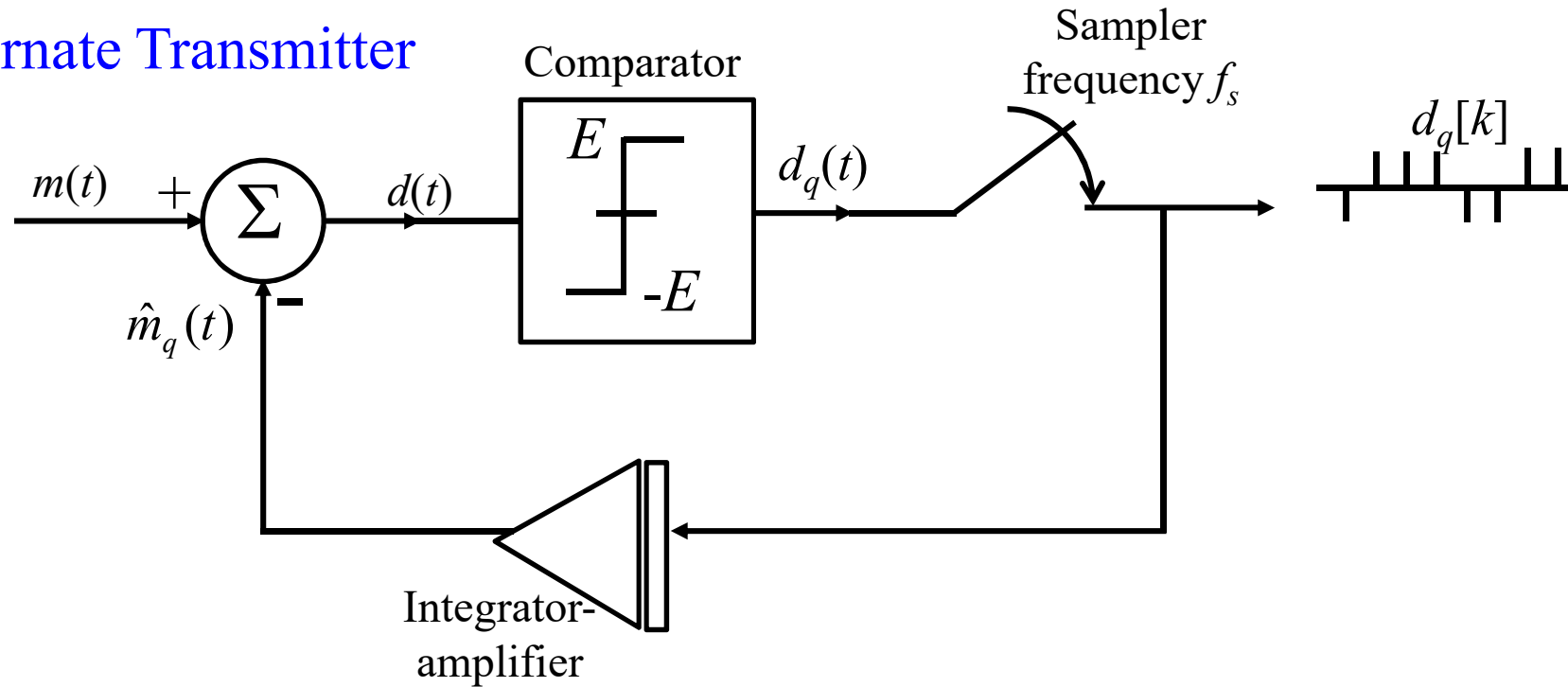
Alternate Transmitter



- Input $m(t)$ is analog
- $d(t)$ is compared in *comparator*: $d(t) > 0$? $d_q(t) = E : d_q(t) = -E$
- $d_q(t)$ is sampled @ f_s to generate $d_q[k]$
- $d_q[k]$'s are integrated and amplified to get $\hat{m}_q(t)$

Delta Modulation

Alternate Transmitter



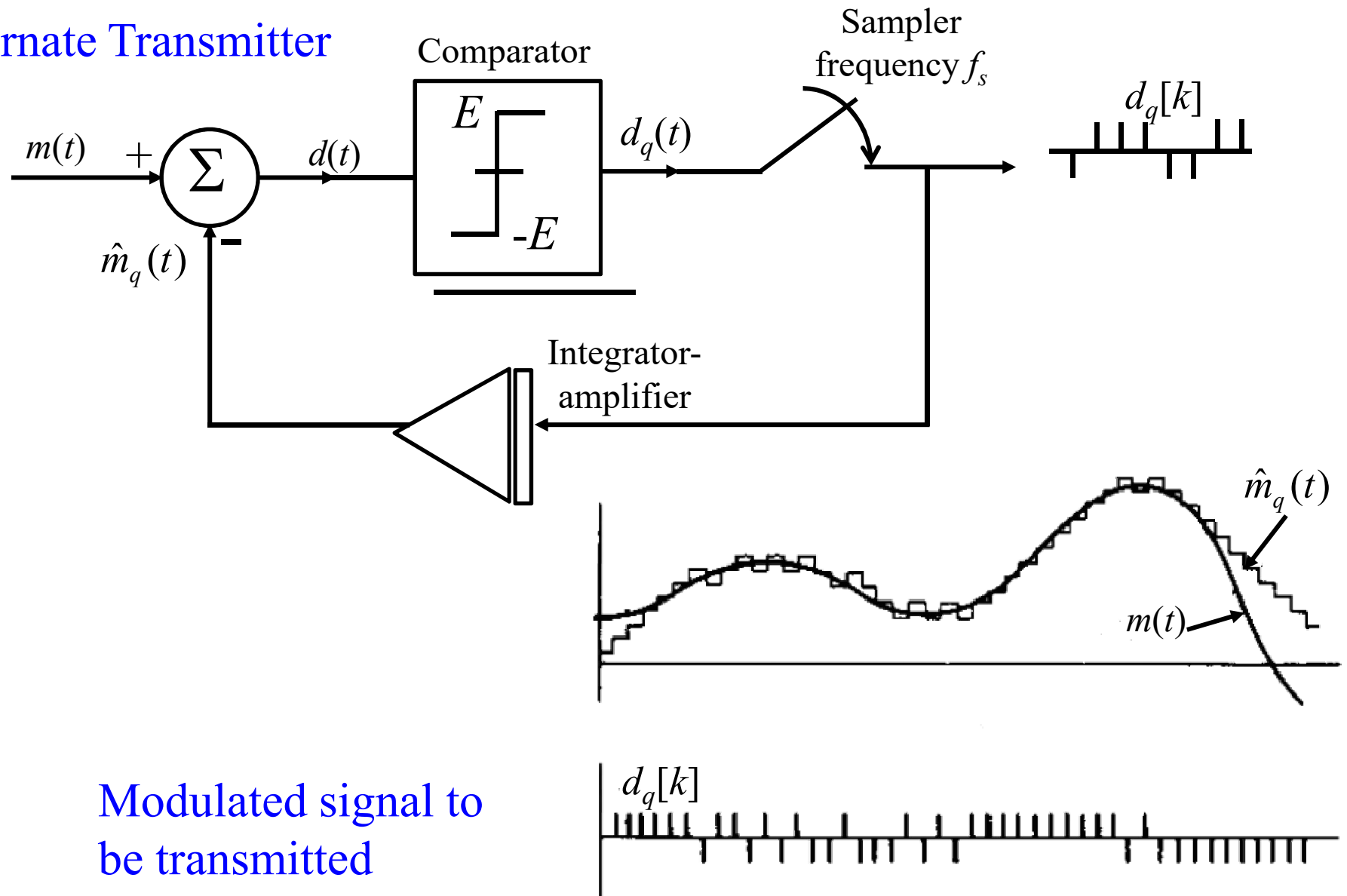
- The sampler generates pulse train $d_q[k]$ as follows

positive pulse if $m(t) > \hat{m}_q(t)$

negative pulse if $m(t) < \hat{m}_q(t)$

Delta Modulation

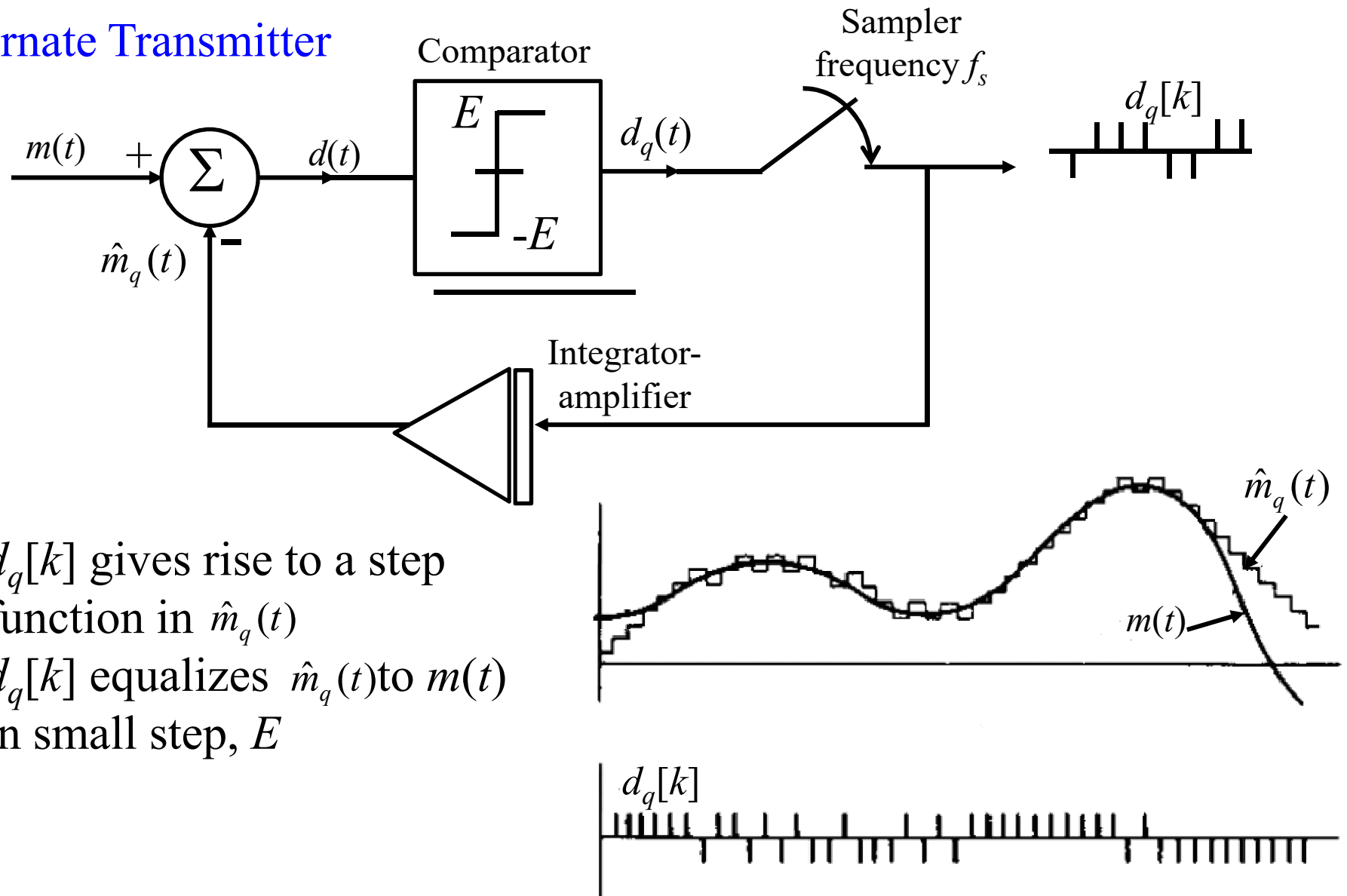
Alternate Transmitter



Modulated signal to
be transmitted

Delta Modulation

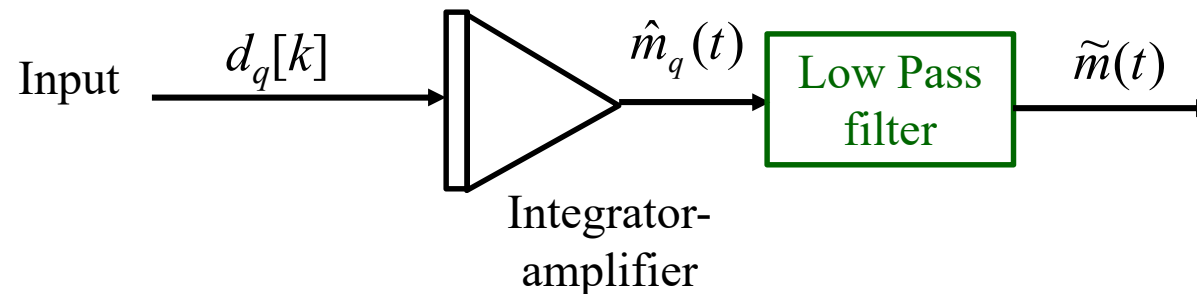
Alternate Transmitter



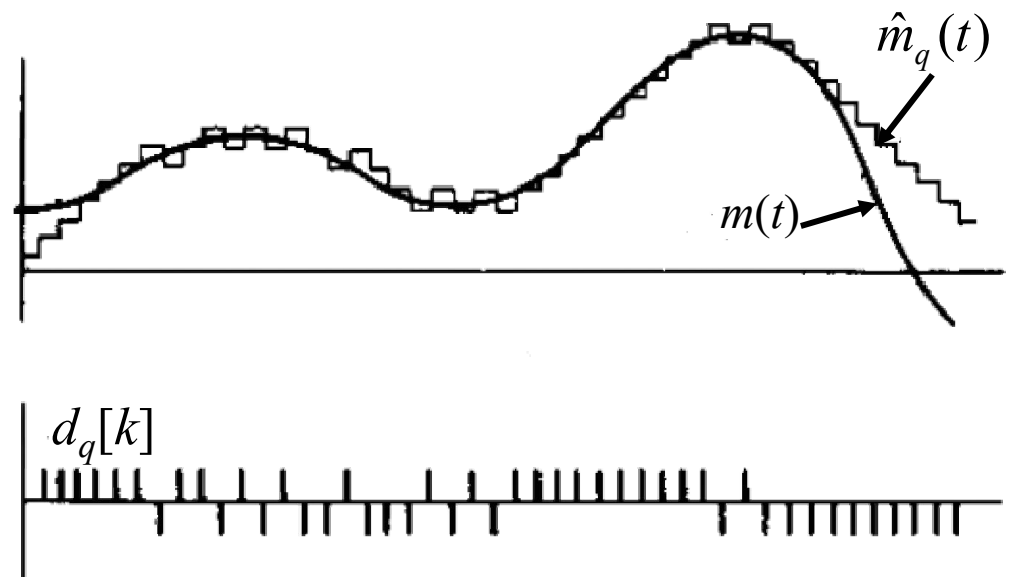
- $d_q[k]$ gives rise to a step function in $\hat{m}_q(t)$
- $d_q[k]$ equalizes $\hat{m}_q(t)$ to $m(t)$ in small step, E

Delta Modulation

Alternate Receiver

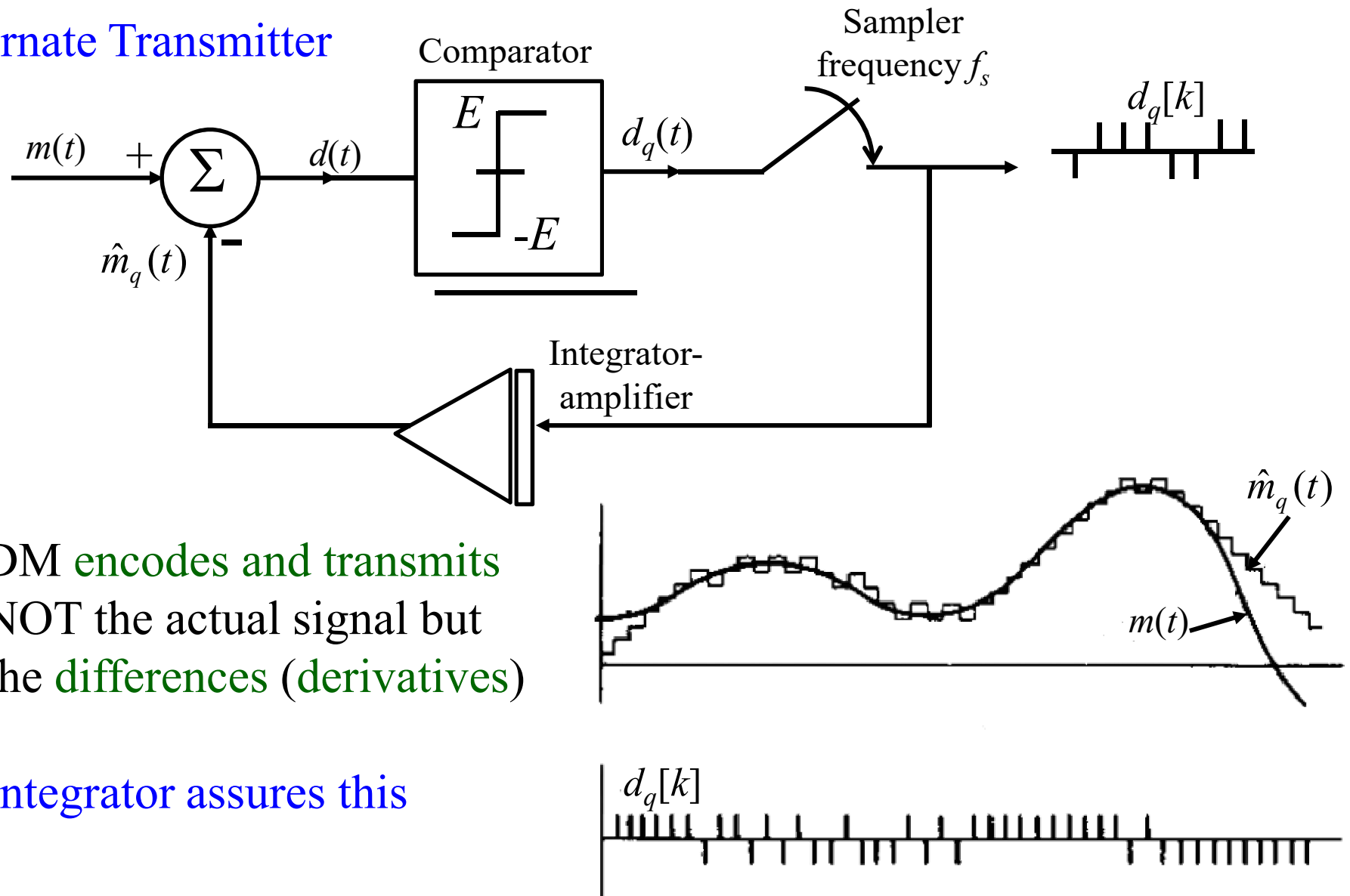


- Coarseness of $\hat{m}_q(t)$ is removed when passed through low pass filter at receiver



Delta Modulation

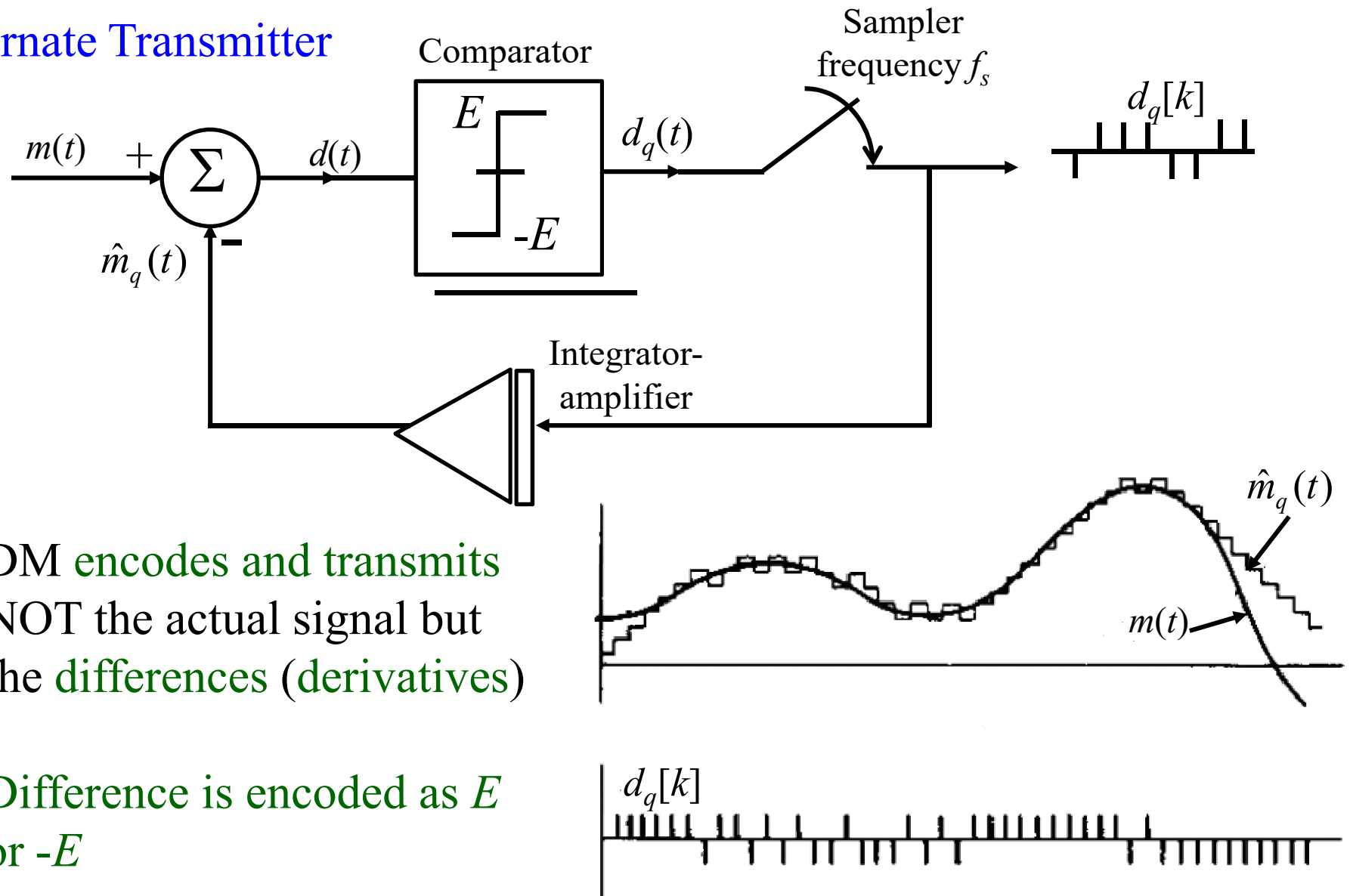
Alternate Transmitter



- DM **encodes and transmits** NOT the actual signal but the **differences (derivatives)**
- Integrator assures this

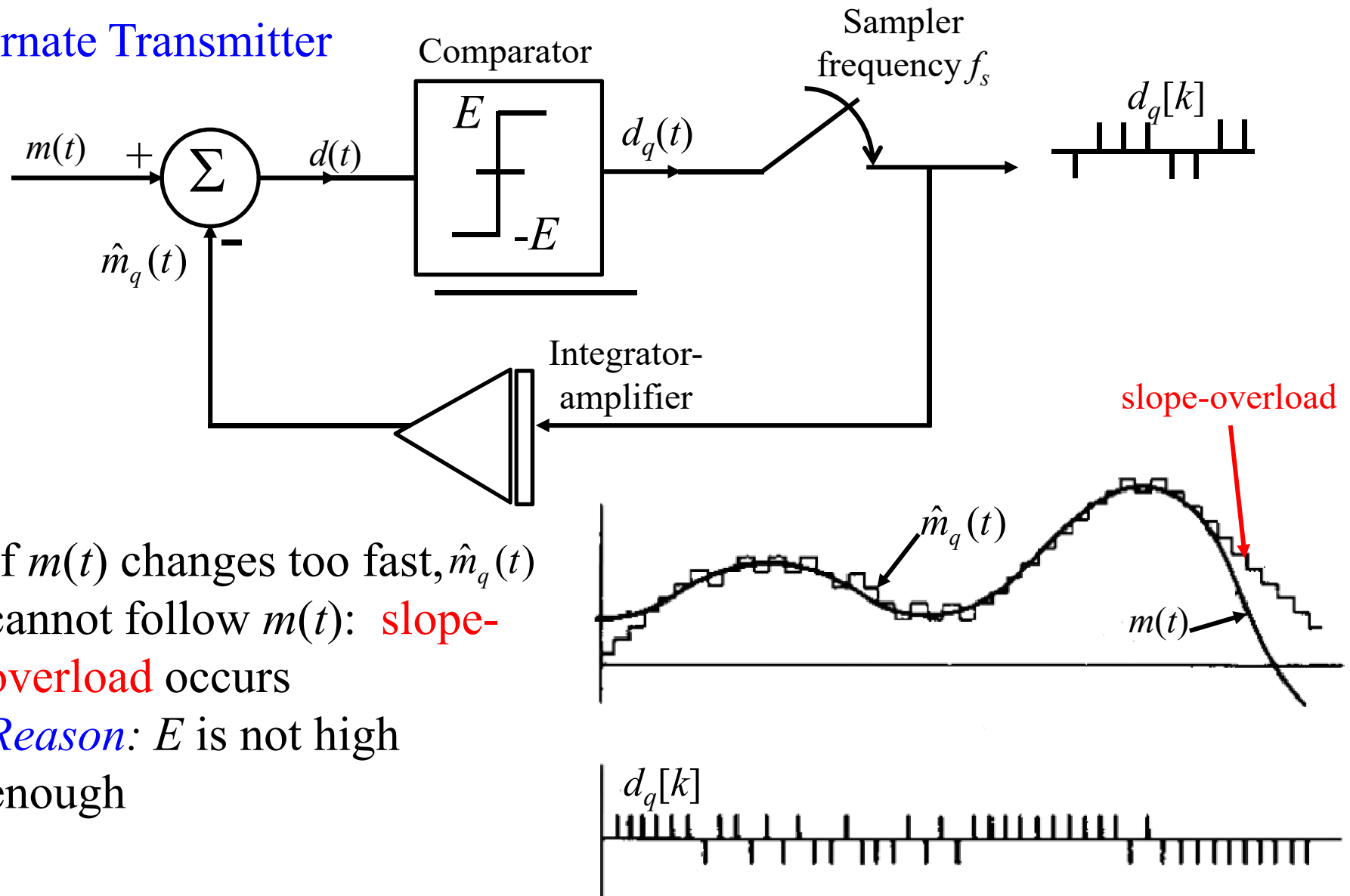
Delta Modulation

Alternate Transmitter



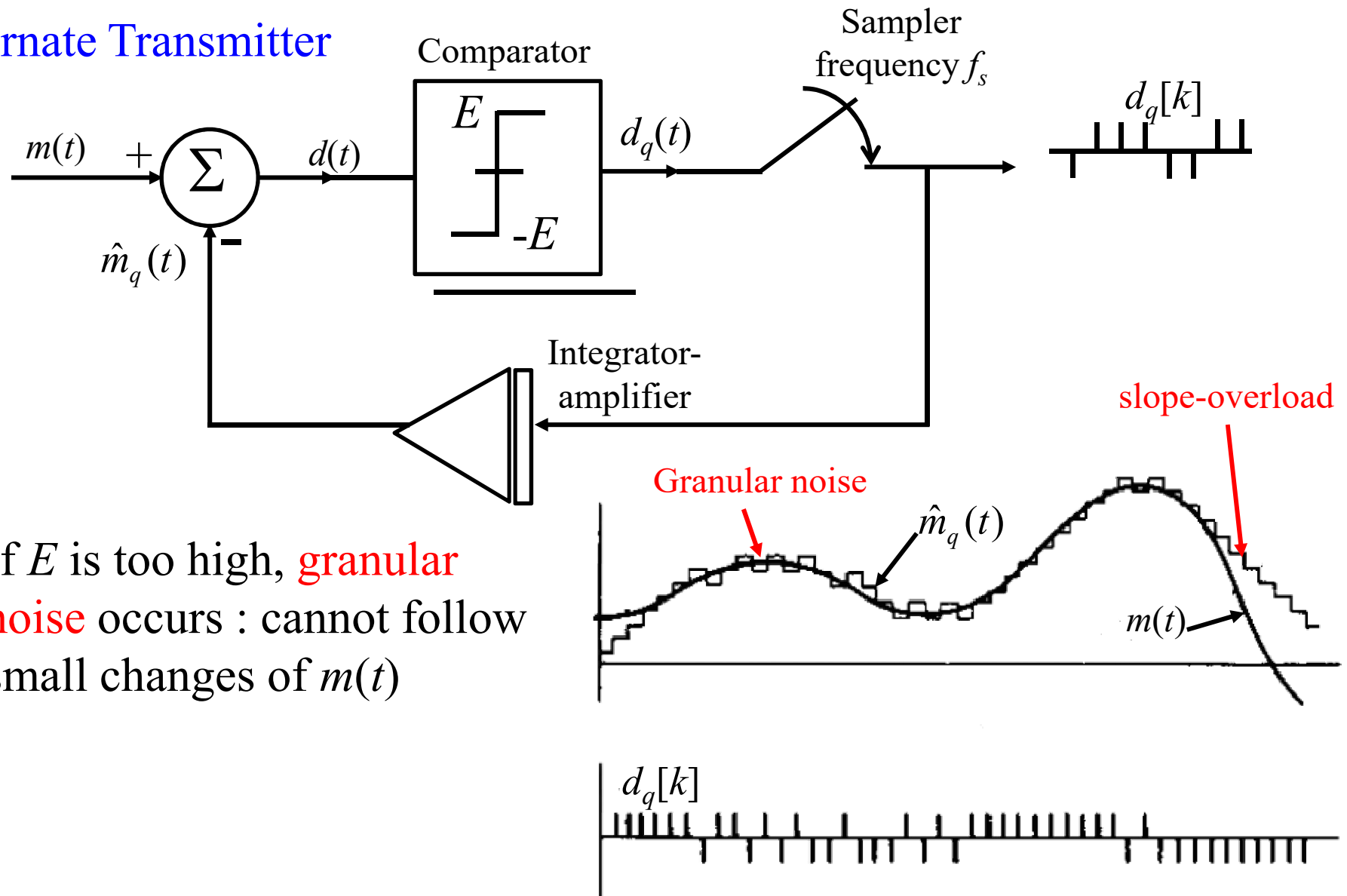
Delta Modulation

Alternate Transmitter



Delta Modulation

Alternate Transmitter



- If E is too high, **granular noise** occurs : cannot follow small changes of $m(t)$